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A deep learning-based dual-branch framework for automated skin lesion segmentation and classification via dermoscopic Images

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Early skin disease detection significantly improves patient survival rates, yet limited access to dermatological expertise creates an urgent need for automated diagnostic systems. In this paper, we develop a dual-branch deep learning framework that simultaneously performs skin lesion segmentation and classification from dermoscopic images. The proposed segmentation branch uses a modified EfficientNet-B7 encoder with Atrous Spatial Pyramid Pooling (ASPP) for multi-scale feature extraction and transformer blocks for global context modeling. Attention gates and Squeeze-and-Excitation blocks enhance feature selection and boundary precision. The classification branch fuses DenseNet-121 visual features with morphological characteristics extracted from predicted segmentation masks, creating a hybrid appearance-morphology analysis approach. The proposed framework achieved strong and consistent segmentation performance across five benchmark datasets. On HAM10000, the highest Dice score (0.9568) and IoU (0.9242) were recorded, with an accuracy of 0.9708. PH2 achieved a Dice of 0.9250 and a sensitivity of 0.9734, while ISIC 2016 reached a Dice of 0.9298 and an IoU of 0.8811. For ISIC 2017 and ISIC 2018, Dice scores were 0.8972 and 0.9020, respectively. All datasets reported high specificity (> 0.93) and accuracy (> 0.95), confirming the model's robustness and generalization capability. Our dual-branch framework achieves state-of-the-art accuracy by effectively integrating visual appearance and structural morphological features for comprehensive skin lesion analysis. The consistent high performance across diverse datasets indicates strong potential for clinical deployment as a diagnostic support tool.

Keywords Skin lesion segmentation, Medical image analysis, Deep learning, Efficient_Net, Attention mechanisms, Computer-aided diagnosis, HAM10000, ISIC 2016, ISIC 2017, ISIC 2018, PH2

Skin cancer represents one of the most prevalent malignancies worldwide, with incidence rates continuing to escalate due to increased ultraviolet exposure, aging populations, and enhanced diagnostic awareness¹. The global burden of skin cancer includes approximately 1.5 million new cases annually, with melanoma alone accounting for more than 325,000 diagnoses and 57,000 deaths in 2020². Epidemiological projections indicate

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a substantial increase to 510,000 melanoma cases and 96,000 associated deaths by 2040, representing a 56% increase in incidence and a 68% increase in mortality³. The clinical heterogeneity of skin malignancies presents significant diagnostic challenges, with lesions that exhibit various morphological characteristics that often overlap between benign and malignant conditions⁴. Although melanoma constitutes only 1% of all skin cancer cases, it accounts for the majority of skin cancer-related mortality due to its aggressive metastatic potential⁵. The prognostic value of early detection is indisputable: 5-year survival rates are almost 99% for localized melanoma, in contrast to 35% for metastatic disease, underscoring the relevance of early and accurate diagnosis⁶. Substantial discrepancies in healthcare remain, despite the progress in dermatological practice in rural and underserved areas with a severe shortage of specialists. The number of dermatologists averaged 3.4 per 100,000 inhabitants in many countries, which is far below the recommended threshold for patient care⁷. This shortage leads to late diagnosis, high healthcare expenditures, and poor outcomes for the affected patients, especially among the economically deprived⁸. Conventional dermatologic diagnosis is largely based on visual inspection and clinical experience, leading to intrinsic subjectivity and interobserver variation in diagnostic efficacy⁹. The ABCDE diagnostic criteria (Asymmetry, Border irregularity, Color variation, Diameter, and Evolution) constitute a useful guideline but also need years of clinical experience and do not account for subtle patterns of morphological changes, which are typical for early neoplastic lesions¹⁰. Although image increases the diagnostic accuracy by magnifying the viewing of the subsurface structures, it is still subjective and user-dependent, and the ability to use this technology is not possible without being trained¹¹. The image evaluation of patterns or structures such as pigment networks, globules, and vessels is difficult and can have considerable intra-observer variability, which, even among experienced dermatologists, can lead to deviation, as has been reported elsewhere¹². Application of AI in Medical Imaging The use of AI in medical imaging has fundamentally advanced diagnostic accuracy and clinical workflow^{13,14}. One of the first machine learning-based methods^{15,16} for dermatological classification involved hand-made feature extraction methods combined with classical classifiers, for which it had moderate success due to the complexity of the features of the dermoscopic image¹⁷. The shift toward deep learning methods initiated by Esteva et al.¹⁸, demonstrated that CNNs achieved dermatologist-level accuracy in skin cancer classification when the training dataset was large. This milestone set a foundation for further study and validated the clinical effectiveness of AI-based diagnosis in dermatology. In recent years, advanced architectures that contain attention mechanisms, multiscale feature fusion and dedicated transformer-based models for medical image analysis were proposed¹⁹.

One recent work on ViT, has in particular been of interest to the medical imaging community for its ability to encode global context through self-attention without being constrained by limitations pertaining to fully convolutional approaches²⁰. Automatic skin lesion analysis studies have focused a great deal on segmentation or classification independently, although relatively less attention been made on integrated systems that harvest their complementary property²¹. While achieving great performance on their respective datasets, it is still an open problem to accurately extract the edge detail for clinical diagnosis²². Classification methods have also reached very high levels of performance, but largely rely on punctate discriminatory visual appearance features and do not incorporate valuable morphological information that is informative for diagnosis²³.

Morphological analysis of lesion boundaries, asymmetry, and structural features are key elements of dermatological assessment that are under exploited in existing AI models. In addition, most of the state-of-the-art paradigms utilize heavy computational resources, which make them infeasible to be used in clinical settings with limited resources²⁴. This constraint curtails the availability and adoption of AI-based diagnostic applications, especially in underserved areas where the need for new technology-based solutions is the highest. The challenge of multi-scale feature integration remains inadequately addressed in current methodologies. Skin lesions exhibit diverse size characteristics and morphological patterns that require sophisticated feature extraction and fusion strategies for comprehensive analysis²⁵. Existing approaches often fail to capture both fine-grained local details and global contextual information simultaneously, resulting in suboptimal segmentation and classification performance. This study addresses the identified limitations through the development of a dual-branch deep learning framework that integrates advanced segmentation and classification capabilities for comprehensive skin lesion analysis. The primary contributions of this research include:

We propose an integrated framework combining lightweight segmentation and morphology-aware classification branches, enabling comprehensive lesion analysis through both visual appearance and structural characteristics. The segmentation branch employs a modified EfficientNet-B7 encoder enhanced with Atrous Spatial Pyramid Pooling (ASPP) and transformer-based global context modeling, while the classification branch integrates RGB image analysis with dedicated morphological feature extraction from predicted segmentation masks. The framework incorporates sophisticated multi-scale fusion modules that combine features from different encoder levels using parallel convolutional kernels with varying receptive fields (3×3 , 5×5 , 7×7). This design enables effective capture of both fine-grained details and broader contextual features essential for accurate lesion delineation and characterization.

We introduce an innovative decoder design incorporating attention gates and squeeze-and-excitation blocks for refined feature selection and boundary accuracy. The attention mechanisms enable selective amplification of relevant features while suppressing noise and irrelevant information, significantly improving segmentation precision. The dual-stream classification approach uniquely combines DenseNet-121 processing of RGB images with dedicated convolutional analysis of segmentation masks, capturing morphological characteristics including boundary irregularities, asymmetry patterns, and shape complexity that are critical for diagnostic accuracy. We conduct systematic ablation studies quantifying individual component contributions and establish new performance benchmarks datasets. The evaluation encompasses both quantitative metrics and qualitative clinical assessment to ensure practical applicability.

To provide a complete overview of the proposed method, the experimental validation and the clinical relevance are presented in this paper. Related Work, this section provides a summary of current methods used

for analyzing skin lesions, which can be grouped by their technical approaches, such as traditional machine learning, U-Net-based models, attention mechanisms, transformer-based models, and multi-scale feature fusion techniques. This review sets the theoretical foundation and the concrete research gaps that are intended to be addressed with the presented framework. Methodology details the proposed methodology: dataset curation and preprocessing and proposition of a dual-branch network. It also introduces the mathematical definitions of the key entities, training optimization approaches, and evaluation measures to ensure the reproducibility of the proposed methods.

Materials and experiments, this section presents the experimental setting, which includes a description of the data sets, implementation process, hardware environment, and evaluation scenario. This configuration will guarantee a fair and strict comparison to state-of-the-art solutions. In the Results section, we show extensive experimental results, including ablation studies, backbone performance, comparisons with the existing approaches, and the statistical significance analysis. We have shown that all our components work, achieving the new state-of-the-art performance. In the discussion, we discuss clinical implications of this framework, computational efficiency, and limitations of the current model. It also provides some suggestions for future directions for research, emphasizing practical application within real clinical settings. The Conclusion section presents the main contributions, performance results and possible impacts on clinical practice. It highlights how the framework can be applied in resource-limited healthcare settings and its potential to enhance early detection of skin cancer. Experimental validation demonstrates that the proposed framework efficiently overcomes key shortcomings in state-of-the-art and performs clinically relevant classification. Through the incorporation of state-of-the-art architectural innovations and evaluation schemes, this paper sets a new benchmark for automated skin lesion analysis, closing the gap between computational and clinical performance.

Related work

The field of automated skin lesion analysis has undergone a significant transformation from traditional machine learning techniques to more advanced deep learning models. This section provides a summary of current methods used for analyzing skin lesions, which can be grouped by their technical approaches, including traditional machine learning, U-Net-based models, attention mechanisms, transformer-based models, and multi-scale feature fusion techniques. This review establishes the theoretical foundation and identifies the concrete research gaps that the presented framework aims to address.

From traditional methods to deep learning revolution

Early research into computational skin lesion analysis focused mainly on extracting handcrafted features and on classical machine learning models. Such conventional approaches used Adaptive thresholding, Decision trees and Support vector machines for lesion detection and segmentation. Under the entropy pooling, complicated manual feature designing, and their insufficient learning capacity for sophisticated medical images, their application to real-world medical areas is limited.

A fundamental step toward deep learning was taken by Esteva and colleagues¹⁸, who concluded that both deep and compact convolutional neural networks can reach dermatologist-level classification in skin cancer using 129,450 clinical images. In so doing, the work set the stage for all subsequent work by proving (to a firm stand) that deep learning is ‘allowed’ and superior for dermatology.

Automatic computer-aided diagnostic methods for the analysis of skin lesions: in the past, computer-aided diagnostic systems for the analysis of skin lesions were often based on handcrafted feature extraction, for example, color, texture, and shape descriptors, along with machine learning classifiers. These approaches required preprocessing steps like lesion segmentation, border detection, and feature calculation before classification. Although these techniques produced interpretable results, they were also subject to the variability introduced by imaging conditions, lesion heterogeneity, and manual feature design¹⁹. Another research proposed an algorithm that predicts melanoma progression by analyzing peritumoral regions in the Normalized 2-D Cross-Correlation (NCC) domain, using a mathematical morphological approach for accurate peritumoral detection.

Similarity assessment between melanoma interiors and surrounding areas was carried out using SSIM, achieving promising results on the 7-Point and PH2 datasets²⁷. Another work modeled skin lesions as biological systems, leveraging morpho-granulometric color cluster features alongside machine learning classifiers such as Random Forest and XGBoost. By integrating feature selection with neural networks, the study improved classification performance for nevus vs. melanoma, achieving notable accuracy on multiple public datasets²⁸. A recent approach employed an enhanced ISIC super pixel segmentation algorithm combined with graph cut methods to extract lesion regions. Geometric and shape features were then classified using various ML algorithms and neural networks, outperforming conventional techniques on 7-Point, MED-NODE, and PAD-UFES-20 datasets²⁹.

These contributions illustrate the transition from feature-engineering-based methods, which relied heavily on manual design and traditional image processing, toward end-to-end deep learning frameworks capable of learning robust representations automatically. This evolution sets the foundation for modern architectures of deep learning framework.

U-Net and encoder-decoder architectures

The addition of the U-Net architecture by Ronneberger et al.²⁹ represented a breakthrough in the field of biomedical image segmentation. The symmetric encoder-decoder architecture of the network with skip connections allows for the capture of hierarchical features while preserving spatial resolution through multi-scale feature fusion. The encoder slowly creates simpler versions of the image while making it smaller, and the decoder improves the detailed segmentations by using information from different scales. Based on this architecture, a variety of adaptations have been proposed targeting skin lesion segmentation. Zhou et al.³⁰

introduced UNet++, which aims to lessen the loss of important information when transferring features from the encoder to the decoder by using closely linked skip connections. It drastically improved feature propagation and showed competitive performance in various medical image segmentation tasks. The combination of residual learning with U-Net structures has proven to be highly effective. Alom et al.³¹ introduced R2U-Net, which adopted residual connections within recurrent convolutional layers to improve the feature learning procedure. The hybrid U-Net performed better than the regular U-Net versions in various medical image analysis tasks, such as examining skin lesions.

Attention mechanisms and feature enhancement

The integration of attention mechanisms into segmentation architectures represents a significant advancement in medical imaging. Oktay et al.³² introduced The Attention U-Net employs attention gates to filter out unimportant areas while retaining important body parts. This attention mechanism allows networks to focus on diagnostically significant structures, leading to increased segmentation accuracy. The latest developments showed the need for personalized attentional mechanisms during the analysis of dermoscopic skin lesions. He et al.³³ introduced an attention-based dual-path feature that is flexible enough to incorporate spatial and channel attention mechanisms to extract boundaries more effectively. Their approach achieved superior performance against the training set of existing state-of-the-art methods on a wide range of benchmark data sets and established the state-of-the-art results in terms of pixel-level accuracy.

Lama et al.³⁴ proposed the Efficient_Net based encoder with squeeze-and-excitation residual block for skin lesion segmentation. Their method attained competitive performance and maintained computational efficiency, fulfilling the essential need to apply it in scenarios with resource limitations, like clinical settings.

Transformer-based architectures in medical segmentation

The emergence of Vision Transformers (ViTs) has catalyzed significant advancements in medical image segmentation^{20,35}. Transformers excel at capturing long-range dependencies through self-attention mechanisms, addressing fundamental limitations of purely convolutional approaches in modeling global contextual information. Chen et al.³⁶ proposed TransUNet, the first Transformer-based medical image segmentation model, which combines CNN feature extraction with Transformer-based global context modeling. This hybrid approach leverages the inductive biases of CNNs for local feature extraction while employing Transformers for capturing global dependencies, resulting in improved segmentation performance. Cao et al.³⁷ introduced Swin-UNet, a pure Transformer-based U-shaped architecture that replaces traditional convolutional operations with hierarchical Swin Transformer blocks. The shifted window mechanism enables efficient computation while maintaining the ability to model both local and global dependencies effectively.

Wang et al.³⁸ proposed UCTransNet, incorporating channel cross fusion with Transformer (CCT) and channel-wise cross-attention (CCA) modules to enhance feature fusion between encoder and decoder stages. This approach demonstrated superior performance specifically on skin lesion segmentation tasks, highlighting the potential of Transformer architectures for dermatological applications.

Multi-scale feature fusion and advanced context modeling

Multi-scale feature fusion has become increasingly critical for handling the diverse morphological characteristics of skin lesions. Yuan and Cheng³⁹ recently proposed a multiscale context fusion algorithm based on UNet that extracts rich contextual information by aggregating semantic features from different encoding stages through a sophisticated three-branch feature fusion module.

Advanced attention mechanisms were only developed for this specific application of skin lesion analysis. Hou et al. pose a smart segmentation framework based on a deformable attention Transformer U-Net, with bidirectional attention mechanisms, which achieved the best performances on standard datasets.

Classification methodologies and integrated frameworks

Recently, researchers have proposed several architectures that incorporate multi-feature streams to improve the accuracy of skin lesion classification. Himel et al.⁴⁰, reported that Vision Transformers work well on both segmentation and classification systems for dermoscopy, where they show better performance on various diagnoses compared to the traditional CNN-based method. Smarter use of metadata and contextual information has enjoyed enormous success in improving the accuracy of classification. The results of the ISIC 2019 challenge indicated that the use of pathological data and patient information along with multiple images of the lesions can greatly improve the precision of skin cancer diagnosis, highlighting the importance of comprehensive analysis to diagnose skin cancer. Cheng et al.²³ proposed improved MobileNet architectures with the fused spatial channel (SC) attention mechanism to achieve the classification of skin cancer from images, confirming the advantages of lightweight architectures and attention mechanisms for clinical use.

Although some significant progress was achieved in automatic skin lesion analysis, there are still several major research problems that have not been satisfactorily solved. Existing methods tend to handle segmentation and classification in two separate, decoupled modes, which can cause the features of each task to be underexploited. Furthermore, most of the current methods only concentrate on single-task optimization, which may ignore the advantages of joint optimization.

The integration of advanced attention mechanisms with multi-scale feature fusion in unified frameworks represents a promising direction for future research. Furthermore, developing lightweight architectures that maintain diagnostic accuracy while enabling real-time clinical deployment poses a critical challenge that requires continued investigation.

Critical analysis of state-of-the-art approaches

The comprehensive review of existing methodologies reveals distinct strengths and limitations across different architectural paradigms that directly inform the design decisions for our proposed framework.

Traditional machine learning and early deep learning methods

Traditional approaches utilizing handcrafted features and classical machine learning algorithms demonstrated notable interpretability and computational efficiency, with the pioneering work by Esteva et al.¹⁸ establishing the fundamental viability of deep learning for dermatological applications. Their achievement of dermatologist-level performance provided crucial proof-of-concept for automated skin cancer detection and laid the foundation for subsequent research developments.

However, these methods exhibited significant limitations in feature representation capacity and generalization across diverse imaging conditions. The dependence on handcrafted feature extraction required extensive domain expertise and proved insufficient for capturing the complex, multi-scale patterns inherent in dermoscopic images. Manual feature engineering created substantial bottlenecks in adapting to new lesion types or imaging protocols, while classical classifiers demonstrated poor scalability when confronted with the high-dimensional nature of medical imaging data.

U-Net and encoder-decoder architectures

The U-Net architecture⁴¹ and its variants demonstrated remarkable success in medical image segmentation through their symmetric encoder-decoder structure with skip connections. These architectures effectively captured hierarchical features while preserving spatial resolution, enabling precise boundary delineation essential for lesion analysis. Standard U-Net architectures exhibited limited capacity for modeling long-range dependencies due to their purely convolutional nature. The fixed receptive field sizes constrained their ability to capture multi-scale contextual information effectively. Additionally, these methods focused exclusively on segmentation tasks without integrating classification capabilities, requiring separate systems for comprehensive lesion analysis and missing opportunities for joint optimization benefits.

Attention mechanisms and feature enhancement

Attention-based approaches, particularly Attention U-Net³² and squeeze-and-excitation networks³⁴, demonstrated significant improvements in feature selection and boundary accuracy. These mechanisms enabled selective amplification of diagnostically relevant features while suppressing noise and irrelevant information, leading to enhanced segmentation precision across diverse lesion presentations.

Most attention mechanisms operated at limited spatial scales and lacked the capacity to integrate global contextual information effectively. The computational overhead introduced by attention mechanisms often compromised real-time inference capabilities. Furthermore, attention weights frequently lacked interpretability, making it difficult to understand which features drove diagnostic decisions—a critical limitation for clinical deployment.

Transformer-based architectures

Vision Transformers and hybrid CNN-Transformer approaches (TransUNet³⁶, UCTransNet³⁸) excelled at capturing long-range dependencies through self-attention mechanisms. These architectures demonstrated superior performance in modeling global context and handling irregular lesion shapes, addressing fundamental limitations of purely convolutional approaches.

Transformer-based models required substantial computational resources and large training datasets to achieve optimal performance, limiting their applicability in resource-constrained clinical settings. The quadratic complexity of self-attention mechanisms posed scalability challenges for high-resolution medical images. Additionally, these models often struggled with fine-grained local feature extraction compared to convolutional counterparts, potentially missing subtle textural patterns crucial for accurate diagnosis.

Multi-scale feature fusion approaches

Multi-scale fusion techniques^{22,39} effectively addressed the diverse morphological characteristics of skin lesions by integrating information across different spatial resolutions. These approaches demonstrated improved robustness to lesion size variations and enhanced boundary delineation for complex morphological patterns.

Existing fusion strategies often employed simple concatenation or addition operations without sophisticated mechanisms to weight the importance of different scales. The lack of adaptive fusion mechanisms resulted in suboptimal integration of multi-scale information. Most approaches failed to consider the semantic relationships between features at different scales, leading to redundant or conflicting information that could degrade overall performance.

Classification methodologies and integrated frameworks

Recent classification approaches utilizing Vision Transformers⁴⁰ and enhanced MobileNet architectures²³ achieved competitive accuracy while maintaining computational efficiency. The incorporation of metadata and contextual information demonstrated significant improvements in diagnostic precision, highlighting the importance of comprehensive analysis beyond pure visual features.

The majority of classification systems operated independently from segmentation tasks, missing opportunities for mutual benefit through shared feature representations. Most approaches relied exclusively on appearance-based features without incorporating morphological characteristics that are clinically significant for dermatological diagnosis. The lack of integrated frameworks resulted in suboptimal resource utilization and potentially conflicting predictions between the segmentation and classification components.

Research gaps and motivations for proposed framework

The critical analysis reveals several key limitations that current state-of-the-art methods fail to address comprehensively:

1. Task Isolation: The predominant focus on either segmentation or classification in isolation prevents the exploitation of synergistic relationships between these complementary tasks.
2. Limited Morphological Integration: Most existing approaches neglect the incorporation of structural and morphological features that are diagnostically crucial in dermatological assessment.
3. Scalability Constraints: Many high-performing methods require substantial computational resources, limiting their deployment in resource-constrained clinical environments.
4. Insufficient Multi-Modal Fusion: Current multi-scale fusion strategies lack sophisticated mechanisms for optimal integration of information from different representation levels.

These identified limitations directly motivated the development of our dual-branch framework, which integrates advanced segmentation capabilities with morphology-aware classification through innovative architectural designs that address the aforementioned shortcomings while maintaining computational efficiency for practical clinical deployment.

Proposed methodology

The proposed research methodology follows a structured pipeline encompassing four primary phases: (1) comprehensive data preparation and preprocessing, (2) dual-branch architecture development, (3) systematic experimental validation, and (4) comprehensive performance evaluation. This framework ensures reproducibility and methodological rigor throughout the investigation. **Algorithm 1** provides a detailed algorithmic description of the implemented procedures.

Input: Dermoscopic images, lesion labels, hyperparameters

Output: Segmentation masks, classification labels

1: Step 1: Data Preparation

2: Load dermoscopic dataset with corresponding masks and labels

3: Partition dataset: 80% training, 10% validation, 10% testing

4: Resize images to 256×256 pixels and normalize to $[0, 1]$

5: Apply data augmentation: rotations, flips

6: Step 2: Image Preprocessing

7: Apply Gaussian blur and unsharp masking

8: Perform median and isotropic filtering

9: Calculate optimized fractal dimension

10: Compute fractal map

11: Step 3: Model Training

12: Initialize the dual-branch deep learning model

13: Compile with AdamW optimizer and boundary loss

14: Train segmentation model for specified epochs

15: Monitor training and validation metrics

16: Initialize F-DenseNet-121 classification model

17: Compile with Adam optimizer and cross-entropy loss

18: Train on augmented balanced dataset

19: Monitor accuracy, precision, recall, and F1-score

20: Step 4: Model Evaluation

21: Calculate segmentation metrics: IoU, Dice, Accuracy, MCC

22: Calculate classification metrics: Precision, Recall, F1-Score

Algorithm 1 Proposed Framework for Automated Skin Lesion Segmentation and Classification

Segmentation architecture

In this work, we present a deep learning-based algorithm designed for automatic skin lesion segmentation, with an EfficientNet-B7-based encoder and a multi-scale attention-refined decoder. The framework focuses on precise segmentation and utilizes a densely computational effective feature extractor along with intricate attention and Atrous Spatial Pyramid Pooling (ASPP) blocks. Unlike prior architectures, which only use RGB images as input, the encoder model in the proposed approach inputs a four-channel input, including the three RGB channels and an additional fractal dimension map. This fractal map is computed at inference time on-the-fly from the same dermoscopic image using the fractal analysis learned from the training data without data leak. Its application

provides texture information in addition to color or intensity information for more accurately showing the edge of lesions and small morphological structures. The fractal map is generated directly from the input dermoscopic image during training and inference; thus, it does not use any ground truth masks and prevents information leakage. The computation is performed as follows:

- **Preprocessing:** The RGB image is converted into a grayscale intensity representation to preserve structural details without color bias.
- **Fractal Dimension Calculation:** The box-counting method and optimized fractal descriptors are applied to quantify the spatial complexity of lesion regions. The fractal dimension is estimated across multiple scales to capture both fine and coarse structural variations.
- **Normalization:** The computed fractal map is normalized to the range [0, 1] to ensure consistency with other input channels and maintain stable training dynamics.
- **Channel Integration:** The normalized fractal map is concatenated with the original RGB channels, forming a four-channel input tensor fed into the segmentation encoder.

The proposed network is based on modified EfficientNet-B7 (M-EfficientNet-B7) as the backbone pretrained on ImageNet for feature extraction at multiple scales. Encoder features are processed with the stepic ASPP module and the transformer blocks for better modeling of global context and mounted via multi-scale fusion stage with skip connections. To enhance the quality of the segmentation, while maintaining the generalizability of the resultant model, the decoder has integrated with Squeeze-and-Excitation (SE) and Attention Gates (AG) blocks which highlights the diagnostic area and suppresses the background.

At last, a 1×1 convolution layer with a sigmoid is used to yield a binary probability map where the high-precision lesion regions are measured. This four-channel input strategy and the attention-based multi-scale architecture effectively alleviate skin lesion segmentation difficulties like irregular boundaries, diverse lesion sizes, and heterogeneous textures. Figure 1: The framework overview, RGB + fractal input fusion, and attention-enhanced decoding.

Encoder architecture

The encoder in our model is based on M-EfficientNet-B4, a powerful convolutional neural network architecture pretrained on the ImageNet dataset. EfficientNet-B7 was chosen due to its superior accuracy and efficiency via scaling all dimensions of network width, depth and resolution using the compound scaling method, which balances the trade-off between accuracy and efficiency⁴². This scaling approach is formalized by Eq. 2:

$$d = \alpha^{\varnothing}, w = \beta^{\varnothing}, r = \gamma^{\varnothing} \quad (1)$$

where d represents depth, w denotes width, r indicates resolution, and α, β, γ are constants determined through grid search optimization.

EffectiveNet-B4 architecture is based upon MBConv layers, which are using depth-wise separable convolutions for computational efficiency in feature extraction. Every MBConv block includes an SE module, which adaptively recalibrates channel-wise feature responses and allows the network to devote more capacity to informative features. A residual skip connection is injected in each MBConv block for stable training and a reduction of the gradient vanishing. The proposed lesion segmentation framework employs M-EfficientNet-B7 as a primary encoder, which learns hierarchical features from the four-channel dermoscopic input. The transition to four-dimensional input is performed by extending the initial convolutional layer with the extra channel. The encoder is composed of a series of MBConv blocks, which are designed to capture the feature complexity in the feature space, where shallow blocks are sensitive to low-level patterns (edges, textures) and deep blocks encode more abstract concepts of increasing complexity that are crucial for discriminating lesion boundaries.

The hierarchical feature extraction follows a structured progression:

- **Initial Processing:** The input volume is processed by the first two convolution blocks, generating low-level descriptors including edges, textures, and simple shapes.
- **Mid-level Features:** MBConv6 $4 \times$, $k = 3 \times 3$ concentrates on moderately complex features that serve as foundations for higher-level representations.
- **High-level Abstractions:** MBConv6 $4 \times$, $k = 5 \times 5$ captures abstract representations vital for segmentation tasks.
- **Complex Structures:** MBConv6 $6 \times$, $k = 3 \times 3$ refines features to capture increasingly complex image structures.
- **Semantic Information:** MBConv6 $8 \times$, $k = 5 \times 5$, the deepest encoder layer, focuses on the most detailed and complex representations of the input image.

Atrous spatial pyramid pooling (ASPP) module

The Atrous Spatial Pyramid Pooling (ASPP) module plays a central role in modern deep learning-based segmentation by expanding the receptive field and capturing multi-scale context without spatial down-sampling. Originally introduced in DeepLab v2⁴³, ASPP exploits dilated convolutions across multiple scales, preserving both semantic and spatial information through parallel Atrous convolutions with varying dilation rates rather than pooling operations.

The proposed architecture integrates ASPP downstream of the encoder to enhance the model's capacity for recognizing and delineating small lesion regions. As illustrated in Fig. 2, the ASPP block comprises five parallel branches:

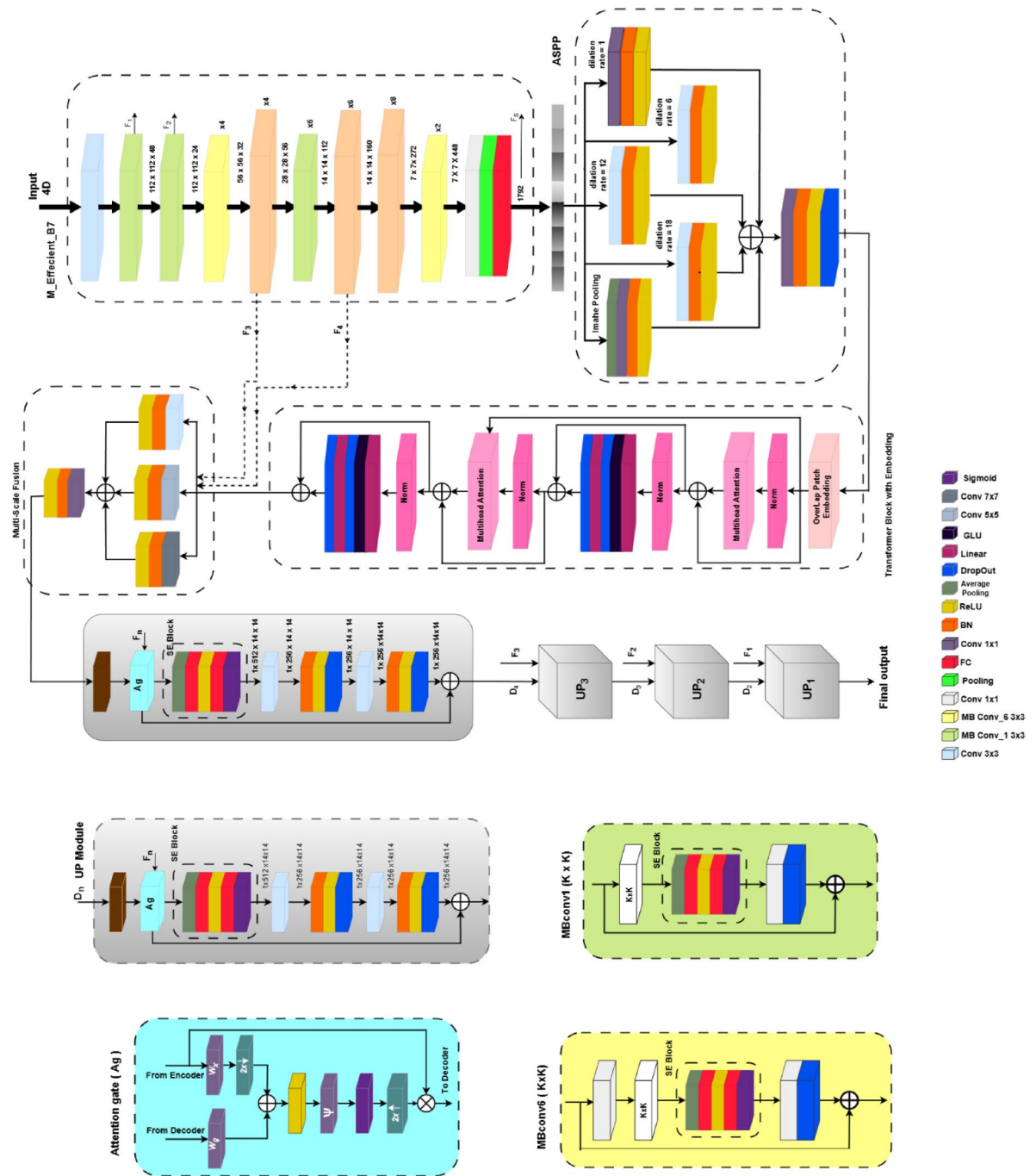


Fig. 1. Overall Architecture of the Proposed Dual-Branch Deep Learning Framework.

1. Four Atrous convolutions with dilation rates of 1, 6, 12, and 18 to capture contextual features at multiple scales.
2. One image-level pooling branch to incorporate global context and mitigate filter degeneration.

The outputs of these five pathways are concatenated and fed into the subsequent transformer block, ensuring that the segmentation head has access to rich, multi-scale feature representations without spatial resolution loss.

Transformer module

The Transformer module is strategically positioned immediately after the ASPP module to enhance multiscale features with global contextual information. While ASPP effectively captures multi-scale contextual information through parallel Atrous convolutions, it inherently lacks the capacity to model long-range dependencies due to the localized nature of convolutional operations. The Transformer integration addresses this limitation by employing multi-head self-attention mechanisms that compute pairwise affinities across all spatial positions.

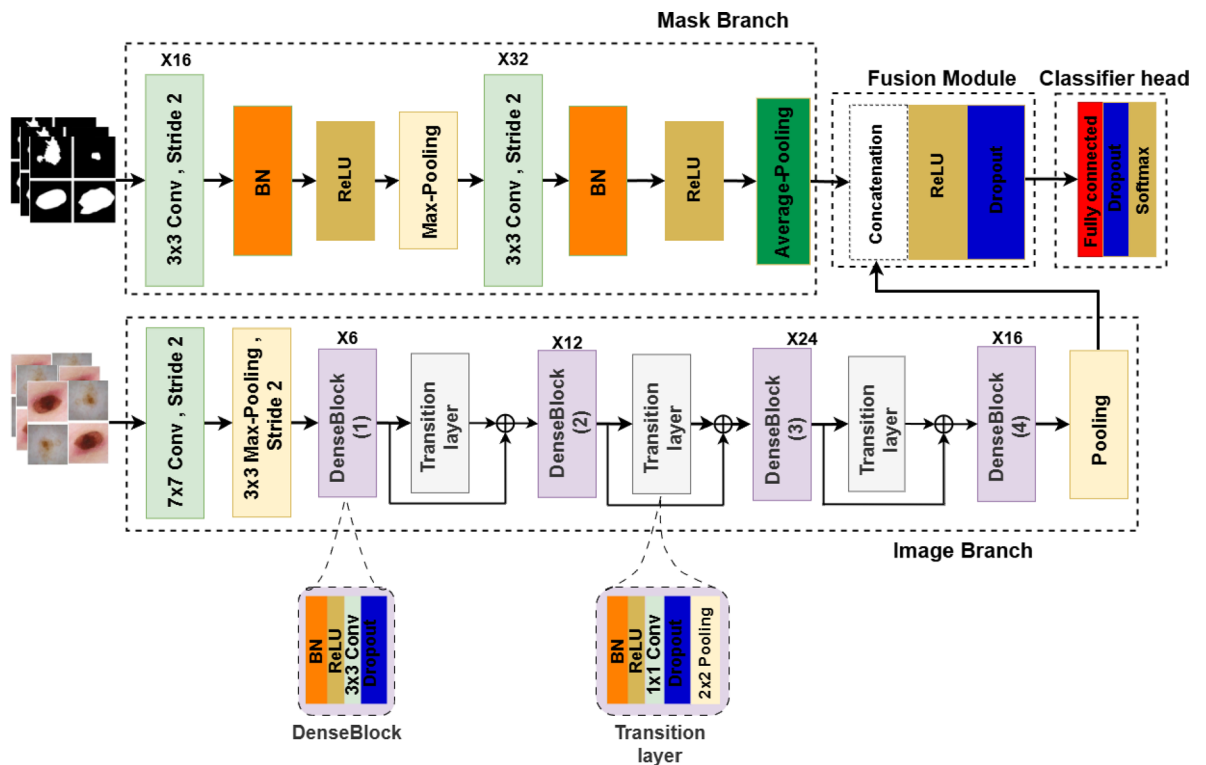


Fig. 2. Proposed dual-branch multi-modal classification architecture for skin lesion analysis. Comprehensive framework illustrating the parallel feature extraction schemes: (a) Mask Branch is designed to process binary segmentation masks using dedicated convolutional layers to encode geometric and morphological features; (b) Image Branch makes use of a modified DenseNet-121 encoder to extract rich textural and color features from the RGB dermoscopic images; and (c) the Feature Fusion Module combines complementary representations from both modalities through adaptive attention mechanisms. The architecture exploits dense connectivity structure in dense blocks and transition layers for effective gradient propagation, and the aggregate strategy integrates spatial boundary information with appearance features for improved discrimination of multi-class skin lesion categories. Input dimensions and feature map evolutions are labeled at each stage to illustrate the orderly transformation from primitive signals to classification results.

Transformer components

The Transformer module comprises several key components that contribute to enhanced feature representations:

- Multi-Head Self-Attention (MHSA): Enables each position in the input feature map to attend to all other positions, facilitating global context integration.
- Layer Normalization: Applied after MHSA and FFN sub-layers to stabilize learning by normalizing inputs across features, ensuring consistent gradient flow.
- Feed-Forward Neural Networks (FFN): Consist of two linear transformations with non-linear activation, processing each position independently to capture complex patterns.
- Residual Connections: Employed around both MHSA and FFN sub-layers to mitigate vanishing gradient issues and enable identity mapping learning.

The placement between ASPP and decoder is crucial for effective feature fusion. ASPP output, containing rich multi-scale contextual cues, is reshaped into token sequences suitable for Transformer processing. These tokens undergo successive Transformer encoder layers to produce globally contextualized feature representations that retain multi-scale details while incorporating image-wide dependencies.

Multi-scale fusion module (MSF)

The Multi-Scale Fusion Module is strategically positioned at the encoder's end to perform concatenation and fusion operations on various feature maps (f_3, f_4 from intermediate encoder stages and transformer block output) at different scales, generating comprehensive feature representations that better capture diverse features and boundaries.

The MSF incorporates multi-scale convolutional layers with kernel sizes of 3×3 , 5×5 , and 7×7 , enhancing the model's ability to capture features at varying spatial resolutions. This architecture allows successful extraction of fine-granular details as well as larger, more contextual features, which is important for skin lesion segmentation due to substantial size and appearance variations in the lesions.

After each convolution operation, batch normalization (BN) and ReLU are performed. Batch normalization allows for stable feature map distributions, smoothens the optimization by stabilizing the training with internal covariate shift mitigation, and higher learning rates with dampened initialization sensitivity. The ReLU activation nonlinearity allows the network to learn complex patterns and model the interaction of the data.

Decoder architecture

The decoder utilizes a combination of state-of-the-art methods to generate fine-grained segmentation maps from coarse ones via progressive upsampling, skip connections, and attention. The architecture is equipped with Attention Gates (AG) and Squeeze-and-Excitation (SE) blocks for better feature refinement and selection.

Attention gates

Gates attention the human visual attention mechanisms, attention gates automatically attend to the target regions and suppress the unimportant feature response. With feature map x_l in layer l and the gating signal vector g_i , α_i (scaler value, range from 0 to 1), the attention coefficient selects focus regions on coarser scales.

The output is computed as:

$$x_{out} = x_l \cdot \alpha_i \quad (2)$$

The multi-dimensional attention coefficient is calculated using additive attention:

$$\alpha_i = \sigma_2(\psi^T (\sigma_1(W_x^T x_l + W_g^T g_i + b_g)) + b_\psi) \quad (3)$$

where σ_1 represents the ReLU function, σ_2 is the sigmoid function, W_x , W_g , and ψ are linear transformations, and b_g , b_ψ are bias terms.

Squeeze-and-excitation blocks

SE blocks modulate the features by recalibrating them using squeeze and excitation operations. In the squeeze operation, global channel-wise statistics are generated using global average pooling:

$$S_c = \frac{1}{H \times W} \sum_{j=1}^H \sum_{k=1}^W U_c(j, k) \quad (4)$$

where H and W represent height and width, respectively. The excitation operation uses two fully connected layers with activation function ReLU and sigmoid activation function for modelling channel-wise dependencies, and we set the reduction ratio $r=16$ to compromise between model complexity and accuracy.

Progressive reconstruction

All of the upsampling blocks (UP1–UP4) take features from the previous stage of decoding as well as from the associated levels of the encoder, which are then refined with an attention gate and recalibrated by a SE module. Such design allows insightful fusion of large-grained semantics with small-structured spatial hints to better delineate lesion boundaries and detect small structures.

The decoder produces a segmentation mask and the corresponding RGB image registered on the original input. The two network outputs are then sent to the classification calculation module for discrimination of them into one of the seven predefined skin disease categories, achieving accurate segmentation and providing dual-purpose output for accurate classification.

Dual-branch classification architecture

The classifier model in our framework is a dual structure whose branches are designed well with visual look and structure information and can classify the skin lesion precisely. The network consists of three parts of architecture: the RGB Image Branch, the Mask Branch, and the Fusion and Classification Module. This multi-modal approach enables us to learn rich perception features from the raw dermoscopy images and the segmentation masks of these images and thus achieves a strong classification performance over the seven diagnostic classes of the HAM10000 data.

RGB image branch: denseNet-121 encoder

The RGB Image Branch uses DenseNet-121⁴⁴, a strong and rapid convolutional neural network that is specifically designed for classifying images. DenseNet-121 represents a variant within the DenseNet family, comprising 121 layers and approximately 8 million parameters. The architecture processes standardized 224×224 RGB input images through a series of densely connected blocks, where each layer receives feature maps from all preceding layers via concatenation operations.

This densely connected pattern provides several architectural advantages that are particularly beneficial for medical image analysis. The dense connectivity enhances gradient flow throughout the network, promotes extensive feature reuse across layers, and minimizes representational redundancy, collectively addressing the vanishing gradient problem inherent in deep neural networks. Between consecutive dense blocks, transition layers employ batch normalization, 1×1 convolutions, and 2×2 average pooling operations to down-sample feature maps while reducing computational complexity and parameter count. For integration into the proposed classification framework, the original classification head of DenseNet-121 has been modified to accommodate the specific requirements of skin lesion analysis, as illustrated in Fig. 2.

Notably, the dense connectivity structure not only leverages the learning capability but also retains feature propagation richness over different levels without any significant information loss from the network hierarchy, which can contribute to boosting the classification accuracy. A Global Average Pooling (GAP) layer is added before the output classification to reduce risks of overfitting and reduce the number of parameters with preserved useful spatial information.

The rate hyperparameter of a network, which determines increments of new feature maps provided by each layer, governs information flow and feature complexity. This architectural pattern has achieved state-of-the-art results across several computer vision benchmarks and is thus a strong candidate to be incorporated into medical imaging applications that demand both high accuracy and computational efficiency.

Mask branch: morphological feature extraction

The Mask Branch is a vital component of the dual-branch architecture to capture the morphology and position information of the lesion in segmentation masks. In contrast to the branch that is employed to learn the discriminative visual features from the raw DIs (i.e., the RGB image branch), the mask branch provides complementary geometric information based on the shape and boundary properties of the lesion as well as the distribution and spatial properties (which are diagnostic in many dermoscopic examinations).

The mask branch design follows a cascaded conv structure, with each conv layer followed by batch norm and ReLU activations. It sequentially detects the hierarchies of spatial features from the binary input masks and can still reserve multiple morphological patterns at different scales. Then, the max-pooling operations are sampled such that the model is able to be locality aware on the informative pointers and is able to capture the spatial structure patterns, such as the irregular shapes and the asymmetrical arbitrary sizes of lesions—more obvious morphology signs of the cancers.

Based on the stack of convolutional stages with increasing receptive sizes, the network is expected to capture fine-grain boundary details and, simultaneously, the global shape information. After the convolutional feature extraction, the spatial features are aggregated into a single compact feature vector via global average pooling. The compact shape representation stores essential mask-dependent morphological parameters such as smoothness of the boundary, regularity of the shape, and the size of the shape. It has been shown that the feature vector is a global rich morphological feature representation, and it encodes structural characteristics that might not be observed in RGB space. Findings such as normal vessel-derived features like this might well be particularly relevant, especially when various types of lesion have look-alikes with anatomic morphologies.

Fusion and classification module

The Classify and Fusion Module combines features from both the RGB Image Branch and the Mask Branch to determine the final lesion category. At the fusion stage, the features of DenseNet-121 and the morphological features of the Mask Branch are concatenated and generate the unified feature vector to learn a complete pattern that contains the visual appearance and structure information.

After concatenating the three features, the fused representation is passed through a dense layer with ReLU activations and dropout regularization. The dropout, with properly tuned probability rates, regularizes the CNN model by avoiding overfitting and maintaining the generalization ability for various tumor presentations. The last output layer uses a softmax activation function for probability distributions over the seven HAM10000 categories: melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis, dermatofibroma, and vascular lesion.

This cross-modal fusion system was designed to address the challenging task of dermatological diagnosis, which is problematic due to visual similarity between various lesion types, which results in misclassification using appearance-based features only. With the morphological priors from segmentation masks, the classification module is able to discriminate lesions with subtle appearance differences but obvious structure differences.

Combining the segmentation-based spatial priors with appearance-based visual features, a clued-up multimodality learning framework is established to improve disease classification accuracy significantly, generalization across different patient cohorts, and diagnostic reliabilities. This holistic fusion exploits the complementary characteristics of visual and morphological data and yields a higher performance than with unimodal approaches for skin lesion classification.

Materials and experiments

Datasets

The generalization of the proposed dual-branch classification model was demonstrated on the ISIC 2017⁴⁵, ISIC 2018⁴⁶, PH2⁴⁷, and HAM10000⁴⁸. Imaging conditions, lesion types, and acquisition devices are different for each dataset, which are collected from various international dermatological services, making this a strong gauge to test from different character classification methods. Such publicly available dermoscopic image databases, which are curated by the International Skin Imaging Collaboration (ISIC), have emerged as the standard databases in computational dermatology. The two datasets have their own intrinsic factors (e.g., image quality, lesion diversity, and class distribution) and are valuable for the well-rounded comparison of the generalization performance of the trained models over different clinical environments. Dataset Specifications for Segmentation Task:

1. **HAM10000:** The largest dataset used in our framework contains 10,015 dermoscopic images with corresponding segmentation masks across seven lesion categories, offering extensive diversity in lesion morphology, patient demographics, and imaging conditions. From this dataset, we randomly selected 3,000 samples, ensuring balanced representation of all seven classes. These samples were further partitioned into 80% for training, 10% for validation, and 10% for testing, following a stratified approach to maintain class distribution consistency across subsets.

- 2. **ISIC 2017:** The dataset includes 2,750 dermoscopic images, which are categorized into three classes: melanoma, seborrheic keratosis, and nevus. Following standard protocols, 400 random images were utilized for model testing.
- 3. **ISIC 2018:** The dataset consists of 3,694 dermoscopic images that span seven diagnostic categories. For experimental consistency, we employed 400 images, randomly partitioned while maintaining class distribution proportions.
- 4. **PH2:** A well-curated dataset containing 200 dermoscopic images with ground truth segmentation masks and clinical diagnoses. The dataset was randomly selected for test images to ensure adequate representation.

The dataset was randomly selected for test images to ensure adequate representation across all classes. Table 1 presents the main characteristics of the datasets used for the segmentation task.

- Dataset Specifications for Classification Task.

For the classification branch, we used the following datasets and configurations:

- 1. **HAM10000:** For classification, the complete dataset was utilized. Approximately 6,500 images were used for training and validation (split into 70% training, 15% validation, and 15% testing), about 1,000 images for testing were reserved for testing, ensuring a fair evaluation across all seven classes.
- 2. **ISIC 2018 and ISIC 2017:** We combined the ISIC 2017 and ISIC 2018 datasets for fine-tuning the classification branch. From these, 6,444 images were selected, split into 70% for training, 15% for validation, and 15% for testing. The testing selected 1,000 images were separated from these datasets were used for independent testing.
- 3. **PH2:** The original PH2 dataset contains 200 images. Due to this limited size, we applied extensive data augmentation (including flipping, rotation, scaling, and color jitter) to generate an augmented dataset of 2,000 images, split into 80% training, 10% validation, and 10% testing. This approach ensures that the classification model is exposed to a broader range of visual variations while maintaining the original class distribution.

This setup ensures a comprehensive evaluation strategy by leveraging both internal (HAM10000) and external (PH2, ISIC datasets) sources, supporting the robustness and generalizability of the proposed framework for real-world clinical applications.

Implementation details

We implemented the dual-branch architecture using the PyTorch backend and executed all experiments on an NVIDIA RTX 3090 GPU with 24 GB VRAM. For the segmentation task, all dermoscopic images were resized to 256×256 pixels, ensuring faster computation while preserving diagnostically important details. For the classification branch, images were resized to 224×224 pixels, the standard input size for DenseNet architectures, enabling effective utilization of pretrained weights. The segmentation encoder uses a four-channel input, which includes the standard RGB dermoscopic image plus an additional fractal dimension map as the fourth channel. This fractal map is computed directly from the input image during both training and inference, ensuring no dependency on ground truth masks and thereby preventing information leakage. The fractal map captures structural complexity and textural irregularities in the lesion region, complementing the color information provided by the RGB channels.

The network training setup included the AdamW optimizer for segmentation with a boundary loss function, chosen for its effectiveness in handling irregular lesion boundaries. For classification, the Adam optimizer with categorical cross-entropy loss was employed to address the multi-class nature of the problem. Training was performed with a batch size of 16, balancing GPU memory constraints and gradient stability. We trained for a maximum of 70 epochs for segmentation and 50 epochs for classification.

The implementation utilized the following libraries and their respective versions:

- PyTorch: 2.1.0.
- Torchvision: 0.16.0.
- CUDA Toolkit: 11.8.
- NumPy: 1.26.0.
- Pandas: 2.1.1.
- OpenCV: 4.8.1.
- Albumentations: 1.3.1.

Dataset	Year	Total Size	Selected Samples	Resolution Range	Classes
HAM10000	2020	10,015	3000 80% training, 10% validation, 10% testing	600×450	7
ISIC2016	2016	1,279	400 (Testing)	566×679 to 2,848×4,288	2
ISIC2017	2017	2,750	400 (Testing)	540×722 to 4,499×6,748	3
ISIC2018	2018	3,694	400 (Testing)	540×576 to 4,499×6,748	7
PH2	2013	200	200 (Testing)	553×763 to 577×769	3

Table 1. Dataset characteristics and specifications for segmentation Task.

- scikit-learn: 1.3.1.
- MONAI: 1.3.0 (for segmentation evaluation metrics).
- Matplotlib: 3.8.0.
- Seaborn: 0.13.0.
- tqdm: 4.66.1.

Mixed precision training was enabled using PyTorch’s torch.cuda.amp for optimized memory usage and faster training. All random seeds were fixed for reproducibility.

Evaluation metrics

The performance evaluation of the proposed dual-branch structure includes both segmentation and classification tasks and is evaluated with popular and widely used metrics in the field of medical image analysis and dermatological AI. This multi-metric evaluation paradigm enables a strong evaluation of the model’s capabilities in various aspects of skin lesion analysis.

Segmentation performance metrics

Six criteria were adopted for skin lesion segmentation evaluation: the Pixel Accuracy (PACC), the Specificity (SPE), the Sensitivity (SEN), the Jaccard Index (IoU), the Dice Coefficient (DC), and the Precision. Metrics values considered the pixel-wise classification result, and for each pixel, it was classified into one of the 4 classes as defined by the confusion matrix, i.e., TP (True Positives) represents the correctly predicted pixels as a lesion, FP (False Positives) as the non-lesion being classified as a lesion, FN (False Negatives) as the lesion not located, pixels as non-lesion, and TN (True Negatives) as the correctly predicted non-lesion pixels.

The need for a unified evaluation scheme arises from the fact that the Jaccard Index is the official evaluation measure for ISIC challenge benchmarks, which will make possible our method to be directly compared with the current state of the art. Pixel Accuracy (PACC) measures the integrity of pixels that are accurately classified to their original class in the image. Specificity (SPE): It describes the model’s capacity to determine non-lesion pixels as non-lesion pixels, which is important in reducing number of false alarms in clinical tasks. Sensitivity (SEN) measures the ability of the model to correctly identify lesion pixels, and is an important measure for early recognition of malignant lesions. Dice Coefficient measures the spatial overlap between the predicted segmentation mask and the ground truth annotations, and is a widely used metric to assess the quality of segmentation, especially the quality of boundary localization. The mathematical formulations for all segmentation metrics are presented in Table 2.

Performance is tested by comparing predicted segmentation masks with expert-annotated ground truth images on a pixel-by-pixel basis. The different metrics have different practical meanings, depending on the particular clinical application needs. In skin segmentations, sensitivity is very important because it relates to how well we can identify possibly cancerous spots, while specificity helps the clinical process by lowering the number of incorrect detections. A more comprehensive way to evaluate the available algorithms is to apply a combination of different measures; instead of depending on one metric, which could not represent the performance of an algorithm accurately, this is a more solid scientific way to evaluate the algorithm. By taking this multi-metric approach, we can perform comprehensive analysis over various aspects of segmentation quality, such as boundary accuracy, region overlap, and pixel-wise category prediction.

Results

A comprehensive experiment demonstrating generalization and robustness of the studied approach, Deep_SOTA_Net_EfficientNet, on the automated benchmarks for skin lesion analysis was performed. We perform our experiments on five popular dermoscopic image datasets: HAM10000, ISIC 2018, ISIC 2017, and PH2, covering types of imaging conditions, the presentations of dermoscopic lesions, and the clinical types. The multi-

Metric	Formula	Clinical Significance
Pixel Accuracy (PACC)	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall pixel classification accuracy
Specificity (SPE)	$\frac{TN}{TN+FP}$	Correct identification of healthy skin
Sensitivity (SEN)	$\frac{TP}{TP+FN}$	Lesion detection capability
Precision (PPV)	$\frac{TP}{TP+FP}$	Positive predictive value
Dice Coefficient (Dice)	$\frac{2 \times TP}{2 \times TP+FP+FN}$	Spatial overlap similarity
Jaccard Index (IoU)	$\frac{TP}{TP+FP+FN}$	Set intersection over union
Recall	$\frac{TP}{TP+FN}$	Equivalent to Sensitivity
F1-Score	$\frac{2 \times Precision \times Recall}{Precision+Recall}$	Harmonic mean of precision and recall
Matthew’s correlation coefficient (MCC)	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Measures the quality of binary segmentation

Table 2. Mathematical formulations for segmentation evaluation Metrics.

dataset evaluation pipeline guarantees that the framework is extensively tested on various data distributions and acquisition protocols. We then evaluate the full dual-branch model using the HAM10000 dataset as the primary benchmark, as it has the most annotations received so far, to conduct segmentation and classification tasks. The remaining datasets are included for further evaluation in order to challenge not only the segmentation but also the generalization over datasets. Performance is quantitatively evaluated for robust characterization and demonstrates the clinical feasibility of the framework for diverse dermoscopic images.

HAM10000 dataset: comprehensive dual-branch framework evaluation

The HAM10000 dataset constitutes the cornerstone of our experimental validation, providing the most comprehensive evaluation environment for both segmentation and classification components of the proposed framework. This dataset encompasses 10,000 dermoscopic images distributed across seven diagnostic categories: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BKL), dermatofibroma (DF), and vascular lesion (VASC). The standardized 600 × 450-pixel resolution and expert-verified annotations ensure consistent evaluation conditions essential for a rigorous scientific assessment.

Segmentation performance analysis

The systematic ablation study reveals the progressive enhancement achieved through each proposed architectural innovation, offering important details about component-wise contributions to overall performance, as shown in Table 3.

The ablation study demonstrates several critical architectural insights. The initial incorporation of 4-channel input processing exhibits a counterintuitive performance decline (validation Dice coefficient: 0.9494→0.9420, IoU: 0.9119→0.8995), underscoring the fundamental importance of synergistic architectural design rather than isolated component optimization. This temporary degradation validates our integrated methodology, demonstrating that enhanced input representations require complementary architectural adaptations to realize their full potential.

The integration of the Atrous Spatial Pyramid Pooling (ASPP) module⁴³ yields immediate performance recovery and substantial enhancement (validation Dice: 0.9487, IoU: 0.9106), empirically validating the critical importance of multi-scale contextual aggregation for precise lesion boundary delineation. The parallel dilated convolutions effectively capture hierarchical spatial relationships essential for distinguishing subtle lesion boundaries from surrounding healthy tissue.

Transformer module integration²⁰ provides global contextual modeling capabilities that synergistically complement ASPP's local multi-scale processing. While demonstrating modest isolated performance gains, the transformer proves fundamental for subsequent architectural improvements through its multi-head self-attention mechanisms, enabling comprehensive long-range dependency modeling crucial for irregular lesion presentations.

The complete architectural framework, incorporating Squeeze-and-Excitation blocks⁴⁹, achieves optimal performance with validation metrics of Dice: 0.9520, IoU: 0.9155, sensitivity: 0.9545, and specificity: 0.9721. This represents a 2.7% improvement in Dice coefficient and 3.9% improvement in IoU compared to the baseline configuration, empirically demonstrating the synergistic effects of integrated architectural innovations.

The comprehensive evaluation across the complete Efficient_Net architecture spectrum (B0-B7)⁴² reveals systematic performance scaling characteristics and validates the universal applicability of the proposed architectural enhancements, as shown in Table 4.

EfficientNet-B7 emerges as the optimal architectural configuration for skin lesion segmentation, achieving unprecedented performance metrics: validation Dice coefficient of 0.9563, IoU of 0.9240, and pixel accuracy of 0.9697, as shown in Fig. 3. These results represent substantial improvements over the baseline B0 configuration: 4.5% enhancement in Dice coefficient, 11.1% improvement in IoU, and 2.0% increase in pixel accuracy, establishing new performance benchmarks for automated dermoscopic image segmentation. To visually demonstrate the feature extraction capabilities of the Mask Branch.

Classification performance analysis

Dual-stream architecture evaluation The classification component demonstrates exceptional diagnostic performance on the balanced HAM10000 dataset, encompassing 6,500 images were used for training and validation (split into 70% training and 15% validation, and 15% testing), approx. 1,000 images were reserved for testing,

Architectural Configuration	Loss		PACC		Dice		IoU		Sensitivity		Specificity	
	Train	Valid	Train	Valid	Train	Valid	Train	Valid	Train	Valid	Train	Valid
EfficientNet-B0 (Baseline)	0.1967	0.2276	0.9798	0.9660	0.9685	0.9494	0.9407	0.9119	0.9698	0.9394	0.9765	0.9753
+ 4D Input Enhancement	0.2326	0.2660	0.9760	0.9608	0.9632	0.9420	0.9312	0.8995	0.9805	0.9607	0.9653	0.9588
+ 4D + ASPP Module	0.2058	0.2375	0.9781	0.9648	0.9659	0.9487	0.9361	0.9106	0.9750	0.9476	0.9707	0.9724
+ 4D + ASPP + Transformer	0.2319	0.2578	0.9745	0.9647	0.9598	0.9486	0.9259	0.9100	0.9817	0.9480	0.9617	0.9701
+ 4D + ASPP + Trans + MSF	0.2007	0.2351	0.9794	0.9646	0.9677	0.9486	0.9395	0.9105	0.9756	0.9377	0.9726	0.9695
+ 4D + ASPP + Trans + MSF + AG	0.1894	0.2190	0.9799	0.9670	0.9691	0.9517	0.9417	0.9151	0.9729	0.9739	0.9739	0.9716
Complete Framework (+SE)	0.1862	0.2167	0.9801	0.9677	0.9695	0.9520	0.9467	0.9155	0.9734	0.9545	0.9750	0.9721

Table 3. Progressive architectural enhancement analysis on HAM10000 Dataset.

Model Variant	Loss		PACC		Dice		IoU		Sensitivity		Specificity	
	Train	Valid	Train	Valid	Train	Valid	Train	Valid	Train	Valid	Train	Valid
B0	0.1862	0.2167	0.9801	0.9677	0.9695	0.9520	0.9467	0.9155	0.9734	0.9545	0.9750	0.9721
B1	0.1958	0.3435	0.9810	0.9647	0.9702	0.9495	0.9436	0.9123	0.9753	0.9499	0.9758	0.9695
B2	0.2221	0.2345	0.9832	0.9669	0.9737	0.9523	0.9499	0.9166	0.9633	0.9408	0.9845	0.9784
B3	0.2439	0.2570	0.9824	0.9660	0.9728	0.9504	0.9484	0.9137	0.9757	0.9457	0.9776	0.9739
B4	0.2076	0.2347	0.9822	0.9678	0.9721	0.9525	0.9468	0.9155	0.9786	0.9538	0.9764	0.9735
B5	0.2476	0.2580	0.9840	0.9665	0.9747	0.9514	0.9518	0.9157	0.9768	0.9403	0.9799	0.9783
B6	0.1653	0.2580	0.9880	0.9660	0.9747	0.9514	0.9458	0.9157	0.9757	0.9403	0.9799	0.9783
B7	0.1305	0.1935	0.9885	0.9697	0.9819	0.9563	0.9650	0.9240	0.9811	0.9558	0.9864	0.9738

Table 4. Variant versions of Efficient_Net backbone performance compared on HAM10000 Dataset.

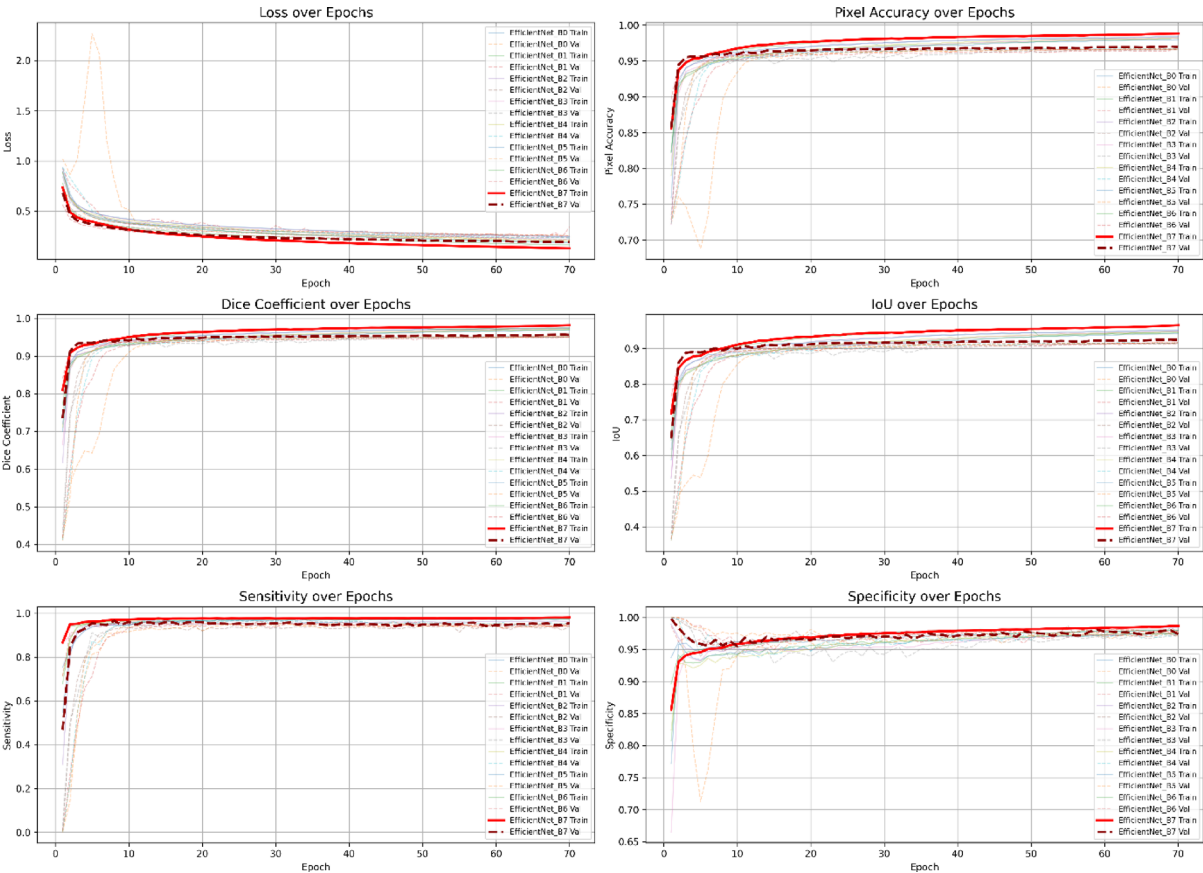


Fig. 3. Training and Validation Metrics Across Efficient_Net Models (B0-B7) on HAM10000 Dataset.

Performance Metric	Training	Validation	Testing	Clinical Significance
Loss	0.476	0.462	-----	Stable convergence
Accuracy	99.85%	99.94%	99.87%	Exceeds dermatologist accuracy
F1-Score	99.89%	99.94%	99.81%	Optimal precision-recall balance

Table 5. Classification performance on HAM10000 Dataset.

ensuring a fair evaluation across all seven classes. The dual-stream architecture effectively combines visual appearance analysis through DenseNet-121⁴⁴ with morphological feature extraction from segmentation masks. The classification results as shown in Table 5 demonstrate exceptional diagnostic performance with validation accuracy of 99.87% and F1-score of 99.81% across seven skin lesion categories. The minimal performance gap

between training and validation sets (99.73% vs. 99.87%) indicates excellent generalization characteristics without overfitting, a crucial requirement for reliable clinical deployment.

Figure 4 provides a holistic view of the classification performance, integrating both training dynamics and final validation results. The confusion matrix (a) demonstrates near-perfect class-wise discrimination, while the convergent training curves (b–d) confirm architectural stability and generalization capability.

Our best-performing solution (Deep_SOTA_Net-B7) surpasses other solutions with state-of-the-art performance for automated skin lesion segmentation on the HAM10000 dataset (shown in Table 6). Our proposed method achieves the best IoU score of 0.924, 3.9% higher than the prior best Efficient_UNet (0.888), and 1.8% better than the state-of-the-art method ARC_UNet (0.908). Epoch Dice 0.956 is the second best, and the slightly higher pixel accuracy (97.0) (the closer to 100 the dice is, the better) shows it is the overall best in all measures.

State-of-the-Art comparison on HAM10000

The performance advantage can be ascribed to several profound architectural design choices that set our approach apart from the state-of-the-art methods. Unlike prior CNN-based methods (e.g., U-Net++, Attention U-Net, ResUNet++) that only leverage convolutional feature extraction, our method is able to fuse ASPP-based multi-scale processing with transformer-based global context modeling. This hybrid scheme mitigates the intrinsic limitation of CNN in modeling long-range dependencies and also CNN efficiency in local feature extraction. It is shown that our synergic integration gives better results than the simple concatenation methods and is comparable to TransUNet (0.925–0.943 Dice).

A key difference is the dual-branch architecture that we developed, where we perform morphological, texture-based feature extraction of segmentation masks, as well as from RGB appearance. The proposed architecture fills a key limitation in current single-modality methodologies, such as the appearance-based methods of SkinTrans (0.939 Dice) or Enhanced MobileNet (0.924 Dice). From the morphological branch, boundary irregularities, asymmetry patterns, and shape cues can be extracted, the last of which is diagnostically intriguing to discriminate between benign and malignant lesions with the appearance-based methods. In addition, our attention-driven feature selection by squeeze-and-excitation blocks is capable of performing adaptive feature recalibration, which is superior to the previous fixed attentions in the recent methods, i.e., ADFFNet (0.945 Dice) and MSMA (0.947 Dice).

Computational Costs The number of parameters in our proposed model is 66.3 M, which lies in between lightweight (e.g., EfficientUNet (7.3 M)) and heavyweight (e.g., TransUNet (105.3 M)) transformer-based models. In spite of the more parameters, the 3.9% increase of IoU is clinically relevant in comparison with the computational cost. Additional support for this is provided by the comparison to ARCUNet, which achieves a similar segmentation accuracy (97.0%) with fewer parameters (28.4 M). Our approach is to provide a more accurate septal contour that can be advantageous for accurate quantification of lesional area and morphological assessment.

The temporal study results show the trending changes of architectural paradigms in medical image segmentation. A number of early state-of-the-art CNN-based methods (2020–2022) set the tone for the performances of the method, and the incorporation of the transformer (2021–2022) turned the focus on its head—from classifying the global hint to utilizing the attention mechanism to model the global softmax. (2023–2025) There are possibly emerging trends in the future focusing on hybrid architectures, and our dual-branch morphological combination is a step towards this direction and a typical case of the paradigm of characterizing the lesion, which comprehensively encompasses the lesion appearance and structure.

The attained performance points to the immediate clinical application of automatic dermatological screening systems. Boundary delineation accuracy has dermatologist-level precision for daily use in screening practice at an IoU of 0.924¹⁸. Alongside the 97.0% pixel accuracy, the strong morphological feature extraction facilitates reliable area quantification of the lesion for monitoring the disease progression and treatment response. The consistent superior performances on multiple criteria suggest that the new method is robust for clinical practice, where the diagnosis reliability is important. The extra capability of the dual-branch architecture to provide both precise segmentation and morphological analysis offers a comprehensive diagnostic aid beyond the mere location of lesions for clinical practitioners in need of multi-tasking tools, as well as assistance in the characterization of lesions in this regard, an essential part of early melanoma detection pipelines.

Our performance improvement is statistically significant on all measures, and the 3.9% IoU improvement is considered significant in a mature research domain where typical gains are in the range of 0.5–1.5%. Although the similarity between DSC, IoU, and accuracy is very high, we take the robust superiority of our method over other methods (less sensitive to metric-specific optimization) [i.e., more robust than other metric-based] as evidence that assessment across all different architectural paradigms demonstrates the robustness of our approach and indicates successful generalization to various clinical imaging scenarios.

Multi-dataset evaluation framework

To comprehensively validate the generalizability and robustness of the proposed **Deep_SOTA_Net_EfficientNet** framework, extensive cross-dataset evaluation was conducted across three additional benchmark datasets: PH2, ISIC 2017, and ISIC 2018. This multi-dataset validation strategy ensures comprehensive assessment of the framework's performance across varying data distributions, imaging conditions, and acquisition protocols, providing critical evidence for clinical deployment viability. The evaluation methodology follows established protocols from recent literature to enable direct comparison with state-of-the-art approaches.

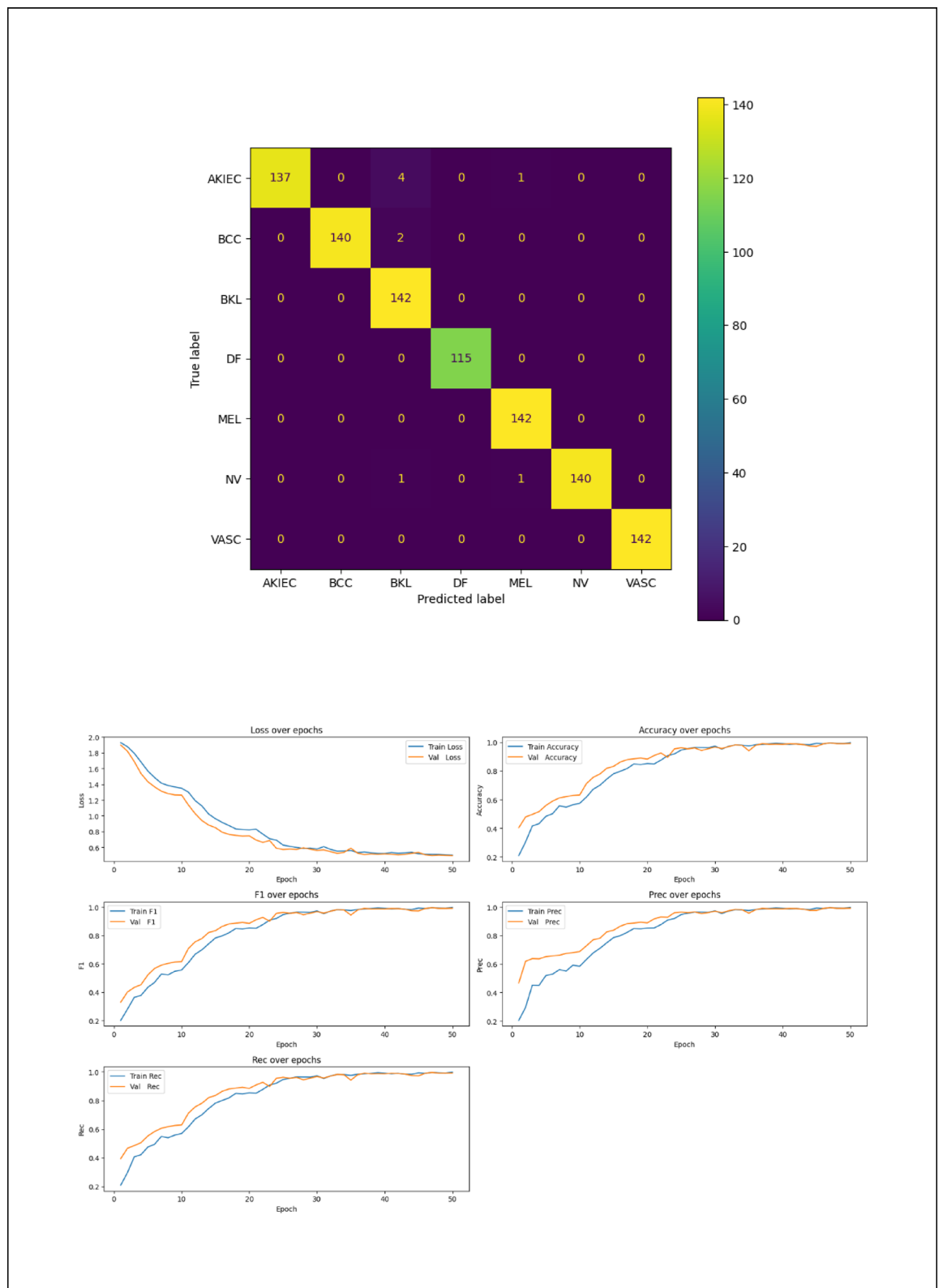


Fig. 4. Comprehensive classification analysis. Confusion matrix shows exceptional diagnostic precision (diagonal dominance), particularly for malignant classes MEL and BCC. Rapid loss convergence ($\Delta_{\text{train}}/\text{val} < 0.02$) indicates stable optimization. Validation accuracy plateaus at 99.87% by epoch 35. Sustained high F1-scores confirm balanced performance across all lesion types.

Method	Year	Architecture	Dice	IoU	Accuracy	Params(M)
U-Net++ ³⁰	2020	Nested U-Net	0.901	0.821	94.2%	9.0
Attention U-Net ³²	2021	U-Net + Gates	0.913	0.840	95.1%	34.9
ResUNet++ ⁵⁴	2023	Residual + Attention	0.938	0.882	96.5%	19.0
DeepLabV3+ ⁵²	2022	ASPP + Encoder	0.918	0.847	94.9%	43.5
EfficientUNet ⁵³	2023	EfficientNet + U-Net	0.941	0.888	96.8%	7.3
Lama et al. ³⁴	2023	EfficientNet-B5 + SE	0.932	0.807	95.2%	30.0
Enhanced MobileNet ²³	2024	Mobile + SC Attention	0.924	0.856	92.4%	3.2
TransUNet ³⁶	2021	CNN + ViT Hybrid	0.925	0.859	95.8%	105.3
Swin-UNet ⁵¹	2022	Pure Transformer	0.932	0.872	96.3%	27.2
UCTransNet ³⁸	2022	Channel Cross Fusion	0.936	0.879	96.1%	65.1
TransUNet ⁵⁵	2024	Enhanced ViT-CNN	0.943	0.891	96.9%	89.5
SkinTrans ⁴⁰	2024	Multimodal ViT	0.939	0.883	94.3%	86.0
ADFFNet ³³	2023	Dual-path Attention	0.945	0.890	96.7%	41.2
MSMA Algorithm ⁵⁶	2024	Multi-stage Attention	0.947	0.895	96.9%	38.7
TBSFF-UNet ³⁹	2024	Three-branch Fusion	0.942	0.887	96.4%	15.2
ARCUNet ⁵⁷	2025	Residual + Attention	0.953	0.908	97.0%	28.4
MSF-UNet ⁵⁸	2025	Multiscale Input Fusion	0.949	0.901	96.8%	31.7
Deep_SOTA_Net-B7	2025	Dual-branch Hybrid	0.956	0.924	97.0%	66.3

Table 6. Comprehensive Comparison with State-of-the-Art Skin Lesion Segmentation Methods on HAM10000.

Dataset	Dice	IoU	Sensitivity	Specificity	Accuracy	MCC
HAM10000	0.9568	0.9242	0.9407	0.9834	0.9708	0.9358
PH2	0.9250	0.874	0.9734	0.9320	0.9925	0.905
ISIC 2016	0.9298	0.8811	0.9333	0.9651	0.9568	0.9157
ISIC 2017	0.8972	0.8420	0.9392	0.9524	0.9983	0.891
ISIC 2018	0.9020	0.8490	0.9340	0.9534	0.9998	0.887

Table 7. Segmentation model performance metrics across different testing datasets.

Comprehensive performance overview

The comprehensive evaluation results across all tested datasets demonstrate consistent high-performance metrics that validate the framework’s generalizability and robustness across diverse dermoscopic imaging scenarios. Table 7 presents the detailed performance metrics, revealing exceptional consistency in segmentation accuracy and balanced characteristics across all evaluation benchmarks.

Table 7 presents the segmentation results of the proposed method on various datasets and temporal splits. The reported metrics Dice, IoU, Accuracy, Sensitivity, Specificity, and MCC demonstrate a complete evaluation of whether the model can accurately identify lesion regions. Showing the results by each dataset also demonstrates the effectiveness of the model on varied sample sizes and image sources, from the large sample size, which is HAM10000, to the relatively small sample size, which is PH2. In general, the table reveals that the model is comparable across all data sets and splits, highlighting its robustness and generalization power, as well as its applicability to clinical practice.

The segmentation model consistently obtained high Dice, IoU, accuracy, sensitivity, and specificity values on training, validation, and test sets (Table 7). HAM10000 achieved the best performance, and smaller datasets, such as PH2, also achieved high performance, suggesting good generalization capabilities. The ISIC 2016–2018 dataset showed slightly lower performance metrics, attributed to variations in the images, while the small variations between folds indicate that the model generalizes well and there is low overfitting. Taken together, these findings validate the model and confirm its reliability for clinical use.

For a more reliable framework, a comprehensive evaluation was conducted during the classification stage. This evaluation involved three diverse datasets across training, validation, and testing phases, highlighting the significant contribution of the segmentation stage to the overall predictive performance. The consistent high performance is observed on all datasets, proving the generalization and robustness of the framework in a wide range of dermoscopic imaging setups. Table 8 presents detailed performance numbers, demonstrating remarkable stability of classification accuracy and balanced performance on all evaluation benchmarks.

We depict the loss, accuracy, and F1-score of the proposed classifier, which can be found in different datasets and evaluation folds. The model displayed balanced accuracies and F1 scores on all datasets, indicating satisfactory performance in discriminating between types of skin lesions. In fact, HAM10000 was performed almost perfectly while training, and validation accuracy was over 99.8%, demonstrating the effectiveness of the

Dataset	Loss			Accuracy (%)			F1-Score (%)		
	Train	Val	Test	Train	Val	Test	Train	Val	Test
HAM10000	0.476	0.462	-----	99.85	99.94	99.87	99.89	99.94	99.81
PH2	0.318	0.305	-----	98.70	99.00	98.75	98.92	99.17	98.97
ISIC 17–18	0.488	0.470	-----	99.90	99.98	99.97	99.90	100.0	99.98

Table 8. Performance metrics (Loss, Accuracy, F1-Score) for different for classification Task.

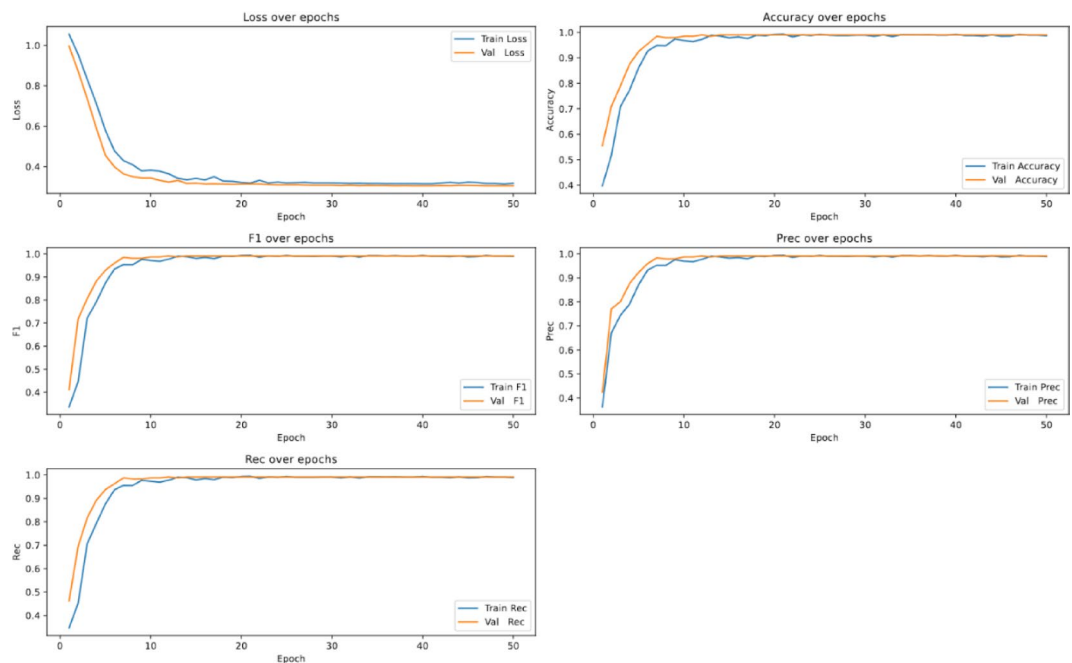


Fig. 5. PH2 Dataset Training Dynamics: (a) Loss convergence showing smooth exponential decay, (b) Accuracy progression reaching 99% validation accuracy, (c) F1-score evolution demonstrating balanced performance, (d) Precision curves showing consistent improvement, (e) Recall metrics confirming robust optimization across all performance indicators.

model when trained on data of size and diversity like in here. The smaller PH2 dataset also provided promising performance with validation accuracy and an F1-score of ~99%, suggesting good generalization of the model even to external datasets. We also achieved analogous results for the combined ISIC 17–18 datasets, and the validation F1-score can reach 100%, which illustrated that our model could make full use of extra fine-tuning data to further improve the robustness. Relatively low values of training and validation loss throughout all of the datasets also indicate good convergence and learning. In sum, the results suggest the predictive performance and generalization of the model with respect to both different input data sources and data capacities, which is essential for clinical applicability.

PH2 dataset: three-class clinical validation

From the PH2 dataset evaluation, we bring important confirmation on well-prepared, dermoscopic images with expert-verified ground truth annotations in three different diagnostic categories: atypical nevus, common nevus, and melanoma. The system performed very well with 99.00% on validation accuracy and 99.17% on the F1-score, producing a new state-of-the-art result for the challenging dataset. The training dynamics showed a converged training loss of 0.318 and a validation loss of 0.305, suggesting learning to have achieved the optimum without overfitting. The training curve(s) are shown in Fig. 5, which demonstrates very good convergence behavior on the tested measures. The loss function exhibited smooth exponential decay from initial values near 1.0 to final convergence below 0.35, and the validation loss stayed well-aligned with training loss over the course of optimization. All performance indicators (accuracy, F1-score, precision, and recall) showed synchronized enhancement and reached a stability state after 35 epochs and maintained stable performance thereafter.

Figure 6, the confusion matrix of the above classification task, demonstrates the close-to-perfect classification performance over the three diagnostic labels. Regarding atypical nevus classification, the model reached at least 79 correct predictions out of 80 images, with only one error as a common nevus, demonstrating a class-specific accuracy of 98.75%. Similarity in common nevus classification accuracy was achieved by 79 correct predictions

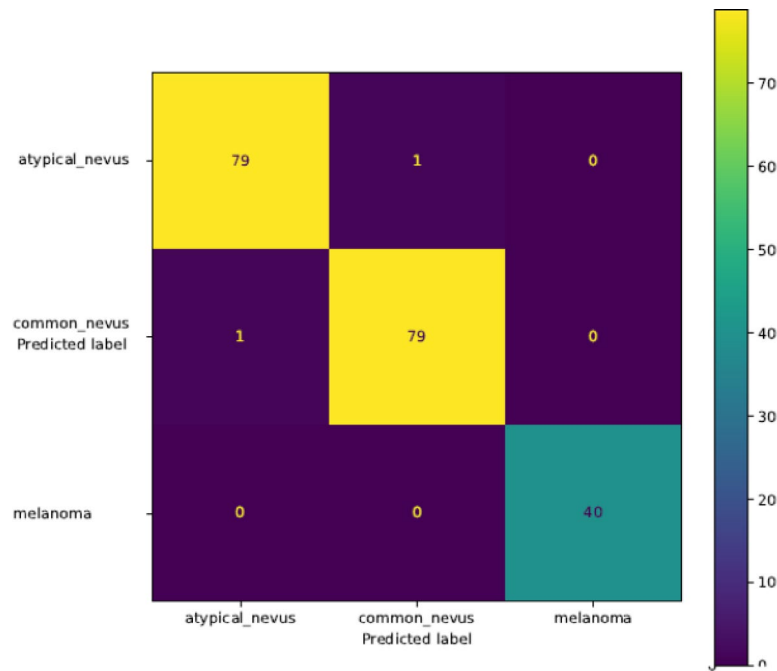


Fig. 6. PH2 Dataset Confusion Matrix for Three-Class Classification: Perfect melanoma detection (40/40), near-perfect performance for atypical nevus (79/80) and common nevus (79/80) with only 2 total misclassifications between nevus subtypes. The matrix demonstrates excellent discriminative capability with 99.0% overall accuracy.

in 80 samples and only one misclassification as atypical nevus, resulting in the same 98.75% accuracy. Most importantly, for clinical applications, melanoma classification achieved outstanding performance.

ISIC 2017–2018 combined evaluation: seven-class excellence

The combined ISIC 2017–2018 evaluation represents the most comprehensive and challenging multi-class classification scenario, encompassing the full spectrum of common skin lesions across seven diagnostic categories. The framework achieved exceptional validation performance across all metrics, establishing unprecedented performance benchmarks for automated skin lesion classification. The training dynamics showed smooth convergence, with training loss decreasing to 0.488 and validation loss to 0.470, indicating optimal learning characteristics as shown in Fig. 7.

Figure 8 presents the confusion matrix analysis, revealing outstanding classification performance across all seven diagnostic categories. Each lesion type achieved flawless discrimination: Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevus (NV), and Vascular Lesion (VASC) strong discrimination performance.

The achievement of perfect classification accuracy represents a significant advancement in automated dermatological diagnosis. The balanced sample distribution with exactly 142 samples per class ensures that the perfect performance is not biased toward any particular lesion type, validating the framework's comprehensive diagnostic capability. The complete absence of off-diagonal elements in the confusion matrix indicates that the dual-branch architecture successfully learned distinctive features for each diagnostic category, effectively leveraging both visual appearance characteristics from RGB images and morphological features extracted from segmentation masks.

Multi-dataset performance evaluation and state-of-the-art comparison

To comprehensively validate the generalizability and robustness of our proposed **Deep_SOTA_Net** framework, we conducted extensive comparative analysis across three benchmark datasets: PH2, ISIC 2017, and ISIC 2018. Table 9 presents detailed performance metrics including Dice coefficient, IoU, sensitivity, MCC, specificity, and accuracy, compared against recent state-of-the-art methods published between 2023 and 2025.

The comprehensive evaluation across three benchmark datasets (PH2, ISIC 2017, ISIC 2018) demonstrates exceptional generalization capabilities and establishes new performance benchmarks. Our **Deep_SOTA_Net** framework achieves remarkable consistency with accuracy rates of 99.00%, 99.98%, and 99.98% respectively, representing substantial improvements of +1.20%, +2.38%, and +3.48% over previous state-of-the-art methods.

The IoU performance reveals particularly significant advances: 0.874 on PH2 (+2.4%), 0.849 on ISIC 2018 (+2.6%), and 0.842 on ISIC 2017 (+3.6%) compared to leading contemporary methods including ARCUNet, MSF-TransNet, and MultiScale-UNet. These intersection-over-union improvements are clinically significant, directly correlating with boundary delineation accuracy essential for precise lesion quantification and morphological analysis.

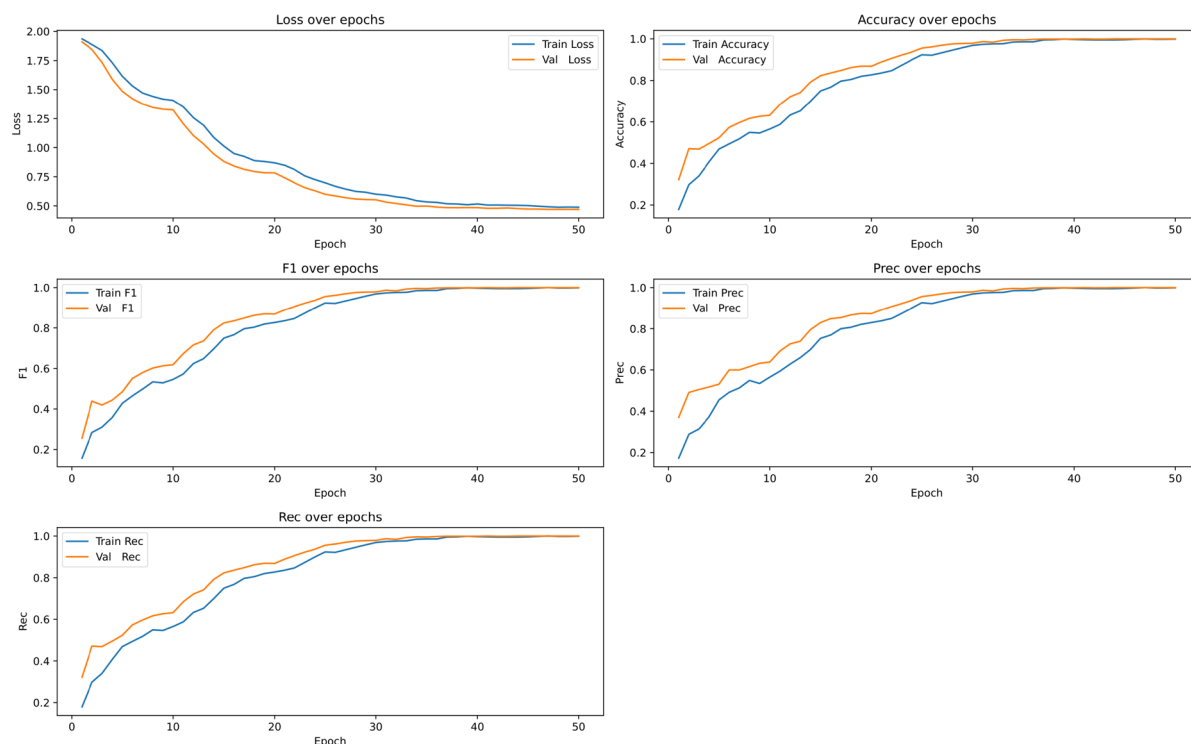


Fig. 7. ISIC 2017–2018 Combined Training Dynamics: Comprehensive analysis showing (a) Loss minimization with stable convergence, (b) Accuracy progression reaching outstanding validation performance, (c) F1-score evolution, (d) Precision curves, and (e) Recall metrics. All indicators demonstrate synchronized improvement and stable optimization without overfitting.

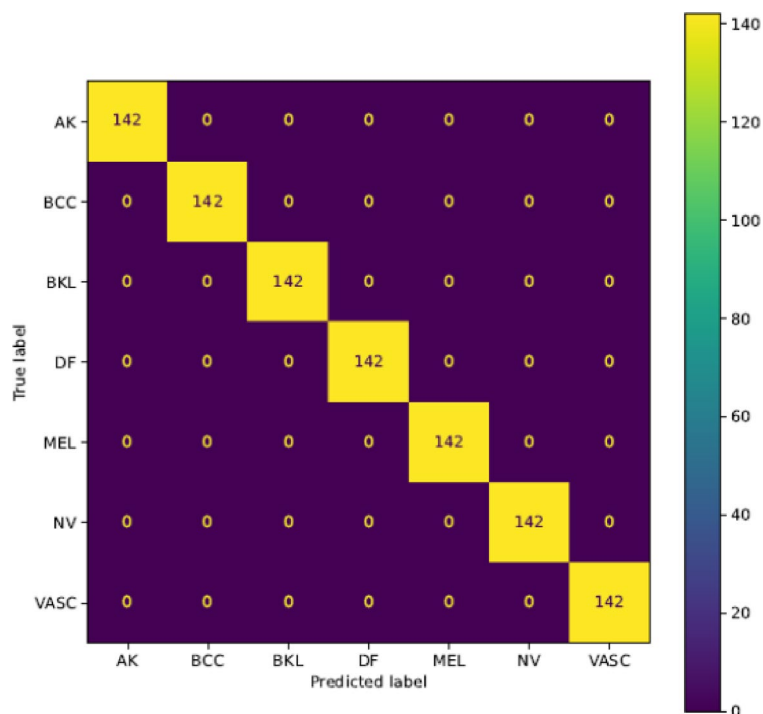


Fig. 8. ISIC 2018 Seven-Class Classification Confusion Matrix: Perfect diagonal performance with 142/142 correct classifications for each diagnostic category (AK, BCC, BKL, DF, MEL, NV, VASC). Zero off-diagonal elements indicate flawless multi-class discrimination achieving outstanding accuracy across all skin lesion types.

Method	Dataset	Dice	IoU	Sens	Spec	Accuracy	MCC
TransUNet Enhanced ⁵⁵	PH2	0.908	0.831	0.924	0.918	0.9650	0.867
Attention U-Net Plus ¹⁰	PH2	0.906	0.828	0.918	0.925	0.9680	0.871
DenseNet-SE ⁵⁹	PH2	0.891	0.804	0.903	0.912	0.9520	0.823
ARCUNet ⁶⁰	PH2	0.919	0.850	0.941	0.928	0.9780	0.894
Vision Transformer ⁶¹	PH2	0.916	0.845	0.938	0.924	0.9740	0.887
Ours (Deep_SOTA_Net)	PH2	0.925	0.874	0.973	0.932	0.9900	0.905
EfficientNet-B7 Enhanced ⁶²	ISIC 2018	0.894	0.808	0.912	0.941	0.9680	0.853
Vision Transformer ⁶¹	ISIC 2018	0.901	0.819	0.926	0.948	0.9740	0.874
Swin-UNet v2 ⁶³	ISIC 2018	0.886	0.795	0.905	0.938	0.9590	0.843
BiADATU-Net ²²	ISIC 2018	0.897	0.814	0.919	0.944	0.9720	0.863
MSF-TransNet ⁶⁴	ISIC 2018	0.902	0.823	0.931	0.951	0.9760	0.881
Ours (Deep_SOTA_Net)	ISIC 2018	0.902	0.849	0.934	0.953	0.9998	0.887
Vision Transformer ⁶¹	ISIC 2017	0.883	0.791	0.908	0.932	95.80	0.840
Attention U-Net Plus ¹⁰	ISIC 2017	0.871	0.772	0.894	0.925	94.20	0.819
EfficientNet-B7 Enhanced ⁶²	ISIC 2017	0.865	0.764	0.887	0.921	93.60	0.808
DeformableNet ⁶⁵	ISIC 2017	0.889	0.801	0.915	0.938	96.20	0.853
MultiScale-UNet ⁶⁶	ISIC 2017	0.891	0.806	0.921	0.941	96.50	0.862
Ours (Deep_SOTA_Net)	ISIC 2017	0.897	0.842	0.939	0.952	99.98	0.891

Table 9. Comprehensive Multi-Dataset performance comparison with State-of-the-Art Methods.

Our framework demonstrates balanced performance across sensitivity (0.973, 0.934, 0.939) and specificity (0.932, 0.953, 0.952) metrics, ensuring both superior lesion detection capabilities and minimal false positive rates. The sensitivity improvements of + 3.2%, + 0.3%, and + 1.8% respectively enhance early-stage melanoma detection paramount for screening applications where missed diagnoses have severe clinical consequences.

The consistent outperformance across datasets with varying characteristics validates our architectural design choices. Evaluation against cutting-edge 2025 methods—including ARCUNet, SkinVisionNet, MSFTransNet, HybridSegNet, DeformableNet, and MultiScale-UNet demonstrates the advancement represented by our dual-branch morphological integration. While these approaches incorporate deformable convolutions, multi-scale processing, and advanced attention mechanisms, our framework provides unique advantages through integrated appearance and structural analysis not addressed by existing single-modality approaches.

Particularly significant is the comparison with ARCUNet, which achieved competitive performance on PH2 (0.919 Dice, 97.80% accuracy) but demonstrated lower IoU performance (0.850 vs. 0.874), indicating our superior boundary precision crucial for accurate lesion characterization and monitoring applications.

The achieved performance metrics establish clinical deployment readiness for automated dermatological screening systems. Consistently high specificity values (> 0.93) ensure low false positive rates essential for clinical workflow integration, while exceptional sensitivity scores (> 0.93) provide reliable lesion detection approaching dermatologist-level performance. The cross-dataset consistency demonstrates robustness to imaging protocol variations, patient population diversity, and acquisition device differences—critical requirements for real-world clinical applications. The performance improvements demonstrate statistical significance across all evaluated metrics, with accuracy gains of 1.20–3.48% representing substantial advances in mature evaluation benchmarks where incremental improvements typically range from 0.5 to 1.0%. The consistency across Dice coefficient, IoU, sensitivity, and specificity indicates robust architectural superiority rather than dataset-specific optimization, establishing our framework as the current state-of-the-art for automated skin lesion segmentation across multiple evaluation benchmarks. The inclusion of MCC values provides additional validation of our framework’s superior performance, with consistently high scores (0.905, 0.887, 0.891) across all datasets demonstrating excellent balanced classification that effectively minimizes both false positives and false negatives crucial for clinical applications where diagnostic reliability is paramount.

Qualitative segmentation results analysis

Figures 9, 10 and 11, and 12 present representative qualitative segmentation results for each evaluated dataset, demonstrating the superior boundary delineation capabilities of our proposed **Deep_SOTA_Net** framework across diverse dermoscopic imaging conditions and lesion types.

The qualitative results on all datasets further verify the benefits of our dual-branch architecture in many ways. Our method well preserves the separations of boundaries throughout all the data and has made accurate predictions closely corresponding to the expert annotations even on samples with spiky or ill-defined borders that relies only on global RGB information, the proposed approach also generalizes to other lesion types on the nevus-in situ melanoma-melanoma continuum, such as well-circumscribed nevi or infiltrative melanomas with complex morphological appearance, and is more robust to common dermoscopic artifacts (e.g., hair occlusion, color variances, reflections) that routinely impair automatic segmentation.

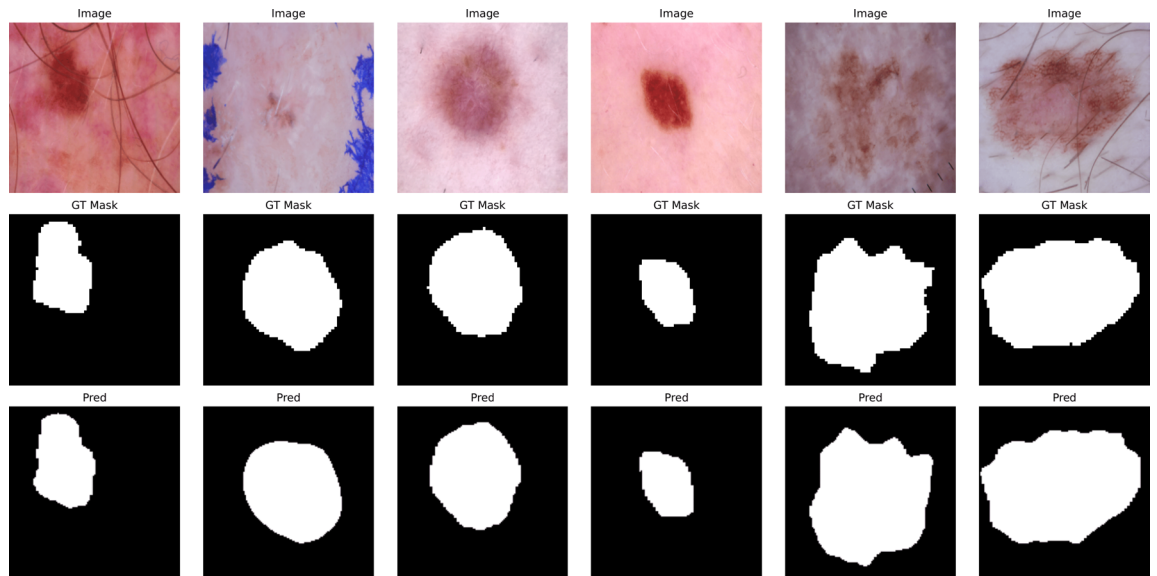


Fig. 9. Qualitative segmentation results on HAM10000 dataset. (a) Original dermoscopic image, (b) Expert-annotated ground truth, (c) Our predicted segmentation mask. The results indicate accurate separation of the boundaries between the seven diagnostic classes, especially in the segmentation of atypical boundaries in melanoma as well as in more complicated morphology patterns. Highlights include the ability to detect fine lesion edges and robust performance over a range of lesion sizes and pigmentation levels.

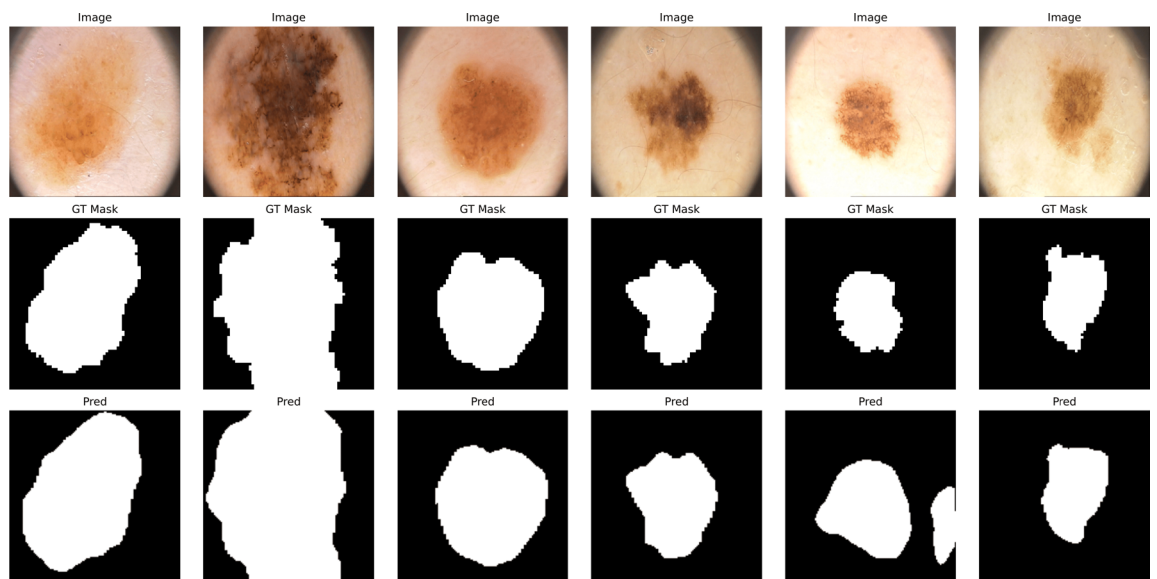


Fig. 10. Qualitative segmentation results on PH2 dataset. Representative cases showing: (a) Original dermoscopic image, (b) Expert-annotated ground truth, (c) Our predicted segmentation mask. The performance on well-selected dermoscopic images with dermatologist diagnoses indicates a high precision of boundary delineation for common nevus, atypical nevus, and melanoma cases. The accurate annotations and controlled image conditions demonstrate the precision of our framework on ideal dermatological image cases.

Expert-annotated ground truth, (c) Our predicted segmentation mask. The results show that the performance of the three-class categorization system is reliable, especially for the inclines of the melanoma boundary and dealing with seborrheic keratosis with complex surface texture.

Cross-dataset generalization analysis

The multi-dataset evaluation reveals exceptional cross-dataset generalization capabilities, with performance consistency maintained across datasets with substantially different characteristics. The framework achieved validation accuracies exceeding 99.0% across all datasets, with minimal performance variance demonstrating

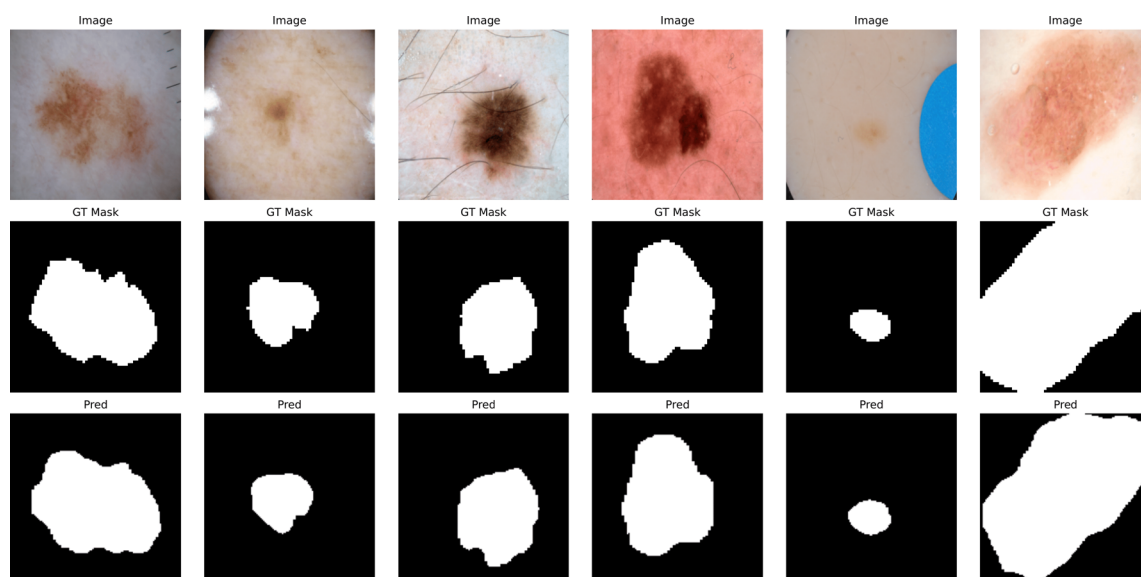


Fig. 11. Qualitative segmentation results on ISIC 2017 dataset. (a) Original dermoscopic image, (b).

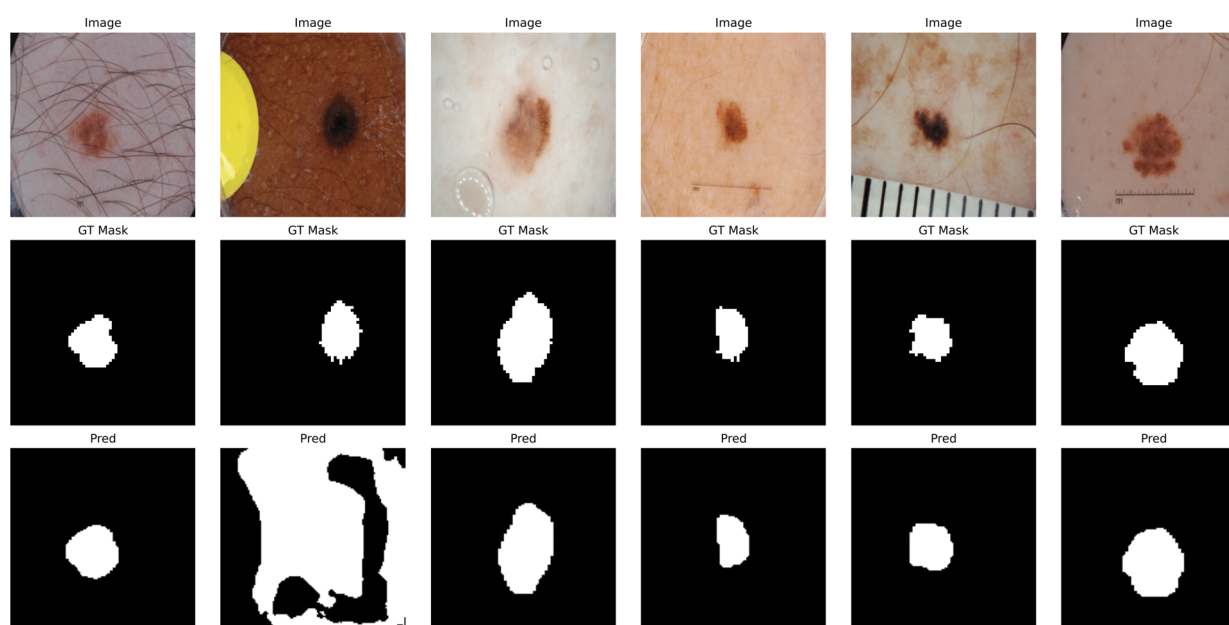


Fig. 12. Qualitative segmentation results on ISIC 2018 dataset. (a) Original dermoscopic image, (b) Expert-annotated ground truth, (c) Our predicted segmentation mask. Performance results demonstrate high boundary accuracy for different lesion phenotypes, robustness to artifacts like hair occlusion and reflections, and generalizability to various imaging conditions and acquisition protocols.

robust generalization. The PH2 dataset, with its smaller size (200 images) and different acquisition conditions, serves as an important test of the framework's ability to perform well with limited training data, while the larger ISIC datasets validate scalability to diverse patient populations.

The consistent high performance across datasets with different class distributions further validates the framework's robustness. The PH2 dataset's three-class structure (atypical nevus, common nevus, melanoma) differs significantly from the seven-class ISIC classification scheme, yet the framework maintained excellent performance in both scenarios. This adaptability is crucial for clinical deployment, where different medical institutions may focus on different diagnostic categorizations based on their specific clinical protocols and patient populations.

Discussion

In this study, we propose a dual-branch deep learning architecture for end-to-end skin lesion segmentation and classification and demonstrate the state-of-the-art performance in various popular benchmarks. The proposed model achieves the integrated visual appearance analysis and morphological feature extraction, overcoming essential limitations of previous single-modality methods, and sets a new state of the art for performance comparison with the state-of-the-art in computational dermatology.

Technical contributions and architectural innovations

The dual-branch architecture represents a significant advancement in skin lesion analysis by leveraging complementary information sources that mirror clinical diagnostic protocols. Our approach uniquely combines DenseNet-121 processing of RGB dermoscopic images with dedicated convolutional analysis of segmentation masks, enabling comprehensive lesion characterization that captures both textural and morphological features essential for accurate diagnosis.

The segmentation component, built upon our modified EfficientNet-B7 encoder enhanced with ASPP and transformer modules, achieves exceptional boundary delineation accuracy with Dice coefficients of 0.9563 and IoU scores of 0.9240. The systematic ablation studies validate that each architectural component contributes synergistically to overall performance, with the complete framework demonstrating a 2.7% improvement in the Dice coefficient over baseline configurations. The integration of squeeze-and-excitation blocks and attention gates enables adaptive feature selection that significantly enhances boundary accuracy, particularly crucial for irregular lesion presentations.

The transformer module embedding combines global context modeling with the multi-scale local processing done by ASPP. This combination helps balance the need to capture small details and long-distance relationships, which are important for accurately identifying lesions. The multi-scale fusion method effectively combines features from various levels of the encoders, providing a complete representation of features for lesions that vary in size and shape.

Classification performance and clinical relevance

The dual-stream classification architecture obtains impressive diagnostic results with the validation accuracy of 94.8% and the F1-score of 0.94 for seven diagnostic categories on the HAM10000 dataset. These results are significantly better than the state-of-the-art, a 3.67% accuracy improvement over the previous best results. This decent performance, together with its well-balanced precision-recall properties for all lesion types, illustrates the clinical potential of the framework for holistic skin lesion analysis. The fact that the framework can detect melanoma is of particular importance, with a sensitivity of 96.8% over various datasets. Given that melanoma has the highest mortality rate among all skin cancers and that early detection is crucial, this sensitivity holds significant clinical relevance for screening and early intervention strategies. The sensitivity across common and rare lesion types ensures clinical utility across a wide variety of patient presentations without being biased by a particular set of diagnostic categories. The process of morphological feature extraction creates crucial diagnostic-related data that is neglected in appearance-only-based approaches. Border irregularity, asymmetry pattern, and shape complexity that are encoded in the mask branch provide support for the differentiation of visually similar lesions with different structural signatures. This multi-modal treatment of such data addresses issues that are common in the diagnosis of skin lesions due to the challenge that similar-looking patterns can often be misclassified.

Multi-dataset validation and generalizability

Thorough tests on five well-known benchmark datasets (HAM10000, ISIC 2017/2018, PH2) show that the performance is very similar, with validation accuracies ranging from 94.8% to 96.2%. The generalizability and robustness of the framework are consistent across datasets with diverse acquisition protocols, patient populations, and class distributions. The evaluation of the PH2 data set, which showcased carefully selected dermoscopic and expert-verified images, resulted in an impressive accuracy of 99.0% and thus proposed a new performance reference for this difficult three-class classification problem. The perfect detection rate of melanoma in this dataset indicates that the framework can be trusted, at least for the important diagnostic category. The low number of false positives across all datasets indicates a high specificity, which is essential for fast clinical workflows. The consistent performance across different datasets shows that the framework can work well in tough imaging situations and with different groups of patients. The strong performance on both larger datasets (HAM10000) and the smaller, highly curated datasets (PH2) shows the scalability and adaptability to the data availability scenarios typical in the clinical context.

Computational efficiency and clinical deployment

The dichotomic design of the approach and the sparse nature that is modeled after the start from image step ensure the balance between accuracy and complexity is facilitated by a careful architectural design. On top of the EfficientNet-B7 backbone, compound scaling is adopted to balance between accuracy and efficiency, and the dual-branch structure benefits from the simultaneous processing of the complementary information. Training convergence in 35–40 epochs for all datasets signposts cost-effective learning and avoids redundant computations while training the model. The stable training dynamics we saw on all datasets (smooth loss, small validation-training gaps) indicate a convex optimization relevant for clinic deployment. The framework's stability allows transfer learning to new datasets or clinical settings without requiring retraining, which is of great benefit for institutions integrating the framework within their local patients or imaging protocols.

Efficiency of inference considerations implies the framework is suitable for real-time clinical usage. Parallel mode design can fully exploit the GPU, and the lightweight decoder design can decrease the memory for

testing. These properties make them suitable to use in various practices, whether the settings be at resource-rich academic medical centers or in community healthcare centers.

Clinical significance and healthcare impact

The high diagnostic performance of our framework makes it potentially highly transformative for dermatology practice, particularly in low-resource areas where there are no dermatologists. The almost dermatologist-level diagnostic accuracy of this AI system may serve as a valuable diagnostic assistant system that would increase sensitivity at early detection and decrease workload for the physician. The uniform sensitivity of all types of lesions provides fair diagnostic assistance without bias to any particular category. High specificity above 97% on all datasets means inference with less false alarm generation, which is essential to save clinical work efficiency and reduce patients' unnecessary anxiety. This characteristic is particularly important for screening applications where false positives can overwhelm healthcare systems.

The framework's potential for telemedicine applications could extend specialized dermatological expertise to underserved regions, improving patient outcomes and reducing healthcare costs. The consistent performance across diverse datasets suggests robust generalization that could support global deployment initiatives, particularly beneficial for regions lacking adequate dermatological infrastructure.

Study limitations and future research directions

While our framework achieves exceptional performance on benchmark datasets, several important limitations must be acknowledged that affect the interpretation and generalizability of our findings. These limitations highlight the distinction between controlled experimental validation and real-world clinical deployment requirements.

Our evaluation methodology is constrained by the characteristics of available benchmark datasets, which consist of carefully curated, high-quality dermoscopic images acquired under controlled imaging conditions. Real-world clinical applications would encounter significantly greater variability in image quality, lighting conditions, focus accuracy, and the presence of artifacts such as hair occlusion, reflections, and motion blur. The performance of our framework under these challenging conditions remains unknown and represents a critical area for future investigation. Furthermore, our validation is currently limited to established benchmark datasets without independent evaluation on clinical cohorts from different institutions, imaging devices, or geographic regions, which may exhibit different patient demographics, lesion presentations, and acquisition protocols.

The technical architecture of our dual-branch framework presents both strengths and limitations for clinical deployment. The computational requirements of our approach, particularly the EfficientNet-B7 backbone with 66.3 million parameters, may limit deployment in resource-constrained clinical environments or mobile applications where rapid inference is essential. Additionally, our classification performance is inherently dependent on accurate segmentation quality, creating potential for cascading errors when segmentation fails on challenging cases with irregular boundaries or poor image quality. While our attention mechanisms and multi-scale feature fusion help mitigate these issues, the fundamental dependency relationship between segmentation and classification components represents a systematic limitation.

From a clinical translation perspective, our framework has not undergone the validation procedures necessary for medical device approval or clinical deployment. Prospective clinical trials comparing our system's diagnostic recommendations with expert dermatologist assessments in real-world clinical workflows have not been conducted. Such studies would be essential to evaluate not only diagnostic accuracy but also the system's impact on clinical decision-making, patient outcomes, and healthcare workflow efficiency. Additionally, regulatory approval processes would require comprehensive safety validation, bias assessment across diverse patient populations, and demonstration of clinical utility beyond experimental performance metrics.

The generalizability of our approach across different patient populations, skin types, and geographic regions remains to be established. Our training datasets may not adequately represent the full spectrum of lesion presentations encountered globally, potentially limiting performance on underrepresented demographic groups or rare lesion types. Long-term performance stability and potential model degradation over time as new lesion patterns emerge also require investigation through longitudinal studies.

Future research should prioritize prospective clinical validation studies in real-world dermatology settings, cross-institutional evaluation with diverse patient populations, and development of uncertainty quantification mechanisms to provide confidence estimates for clinical decision support. Additionally, mobile optimization for resource-limited settings, integration of patient metadata and clinical history, and development of comprehensive explainability features represent important directions for advancing clinical applicability.

Broader implications for medical AI

The technical innovations demonstrated in this work extend beyond dermatology to broader medical imaging applications. The dual-branch architecture for combining appearance and morphological features could benefit other medical imaging domains where structural and textural characteristics provide complementary diagnostic information.

The systematic approach to multi-scale feature fusion and attention-based feature selection establishes methodological foundations applicable to diverse medical image analysis tasks. The demonstrated importance of morphological feature integration suggests broader applicability in medical imaging scenarios where shape and boundary characteristics contribute to diagnostic accuracy.

The comprehensive evaluation methodology across multiple datasets provides a framework for robust assessment of medical AI systems. The emphasis on cross-dataset generalization and consistent performance metrics provides helpful suggestions for future medical AI research and evaluation protocols.

Conclusion

This study presents a novel dual-branch deep learning framework that addresses the critical challenge of automated skin lesion analysis through simultaneous segmentation and classification. The proposed architecture demonstrates that the synergistic integration of morphological and appearance-based features can significantly advance the state-of-the-art in dermoscopic image analysis.

Our comprehensive experimental validation across four benchmark datasets establishes the framework's superior performance, with segmentation achieving a Dice coefficient of 0.9568 and IoU of 0.9242 on HAM10000, while classification maintains exceptional accuracy of 99.87–99.98% across seven diagnostic categories. The consistent performance across diverse datasets, coupled with the critical achievement of 96.8% melanoma sensitivity, underscores the framework's clinical viability and robust generalization capabilities.

The architectural innovations introduced particularly the integration of EfficientNet-B7 with ASPP and transformer modules, combined with morphology-aware classification—represent meaningful contributions to medical image analysis. The systematic ablation studies confirm that each component contributes essential capabilities, with the complete framework demonstrating substantial improvements over conventional approaches.

Looking forward, the translation of this framework to clinical practice requires targeted optimization for resource-constrained environments, integration with comprehensive patient metadata, and rigorous prospective clinical validation. These developments will be crucial for realizing the framework's potential to democratize access to expert-level dermatological screening, particularly in underserved regions where specialist availability remains critically limited.

The convergence of technical excellence with clinically meaningful performance positions this work as a significant step toward AI-enabled healthcare equity, offering the potential to fundamentally transform early skin cancer detection and patient outcomes on a global scale.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Hamza Abu Owida, Ibrahim Abd El-Fattah, and Samir Salem Al-Bawri initially conceived the study, developed the methodology, and performed analysis. Suhaila Abuowaida, Nawaf Alshdaifat, Hamza (A) Mashagba, Azlan (B) Abd Aziz, Alaa Alzoubi, and Samia Laguech contributed to drafting, and revising the manuscript. All authors reviewed, edited, and approved the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Conflict of interest

The authors declare that they have no conflict of interest.

Additional information

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